

NEW ZEALAND CARBON MONITORING SYSTEM

Indigenous Forest and Shrubland Data Collection Manual

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Manaaki Whenua
Landcare Research



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Summary

The New Zealand Carbon Monitoring System aims to advance New Zealand's reporting commitments under the Framework Convention on Climate Change and the Kyoto Protocol, and to provide a framework for national-scale biodiversity reporting. For indigenous forests and shrublands this is achieved by using a grid-based network of permanent plots to provide an unbiased estimate of carbon storage and plant biodiversity. This manual describes the design of the forest and shrubland sampling programme, specifies the data collection and sample processing protocols, and outlines the associated quality control procedures.

Keywords: indigenous forest, shrubland, carbon stocks, Kyoto Protocol, plant biodiversity, vegetation monitoring, permanent plot network, National Vegetation Survey databank.

1 Introduction

The New Zealand Carbon Monitoring System (NZCMS) uses a stock-approach to estimate changes in carbon storage in indigenous forests and in shrublands. Carbon stocks in live biomass and woody debris are estimated by (i) collecting data related to the volumes of trees, shrubs and woody debris from a national network of permanent plots, (ii) converting these volumes to carbon stocks per hectare using regression equations, and (iii) multiplying carbon stock per hectare values for forests and shrublands by the respective areas of these vegetation types obtained from high-resolution satellite imagery. This manual describes the protocols associated with the first of these steps, and describes the protocols for plant biodiversity inventory on the NZCMS plot network.

2 Inventory Design

2.1 Objectives

The primary objective of the NZCMS is to advance New Zealand's reporting commitments under the Framework Convention on Climate Change (FCCC), and the Kyoto Protocol.

The framework being developed for the NZCMS also has the potential to contribute to a range of national and international reporting requirements. Internationally these include initiatives associated with sustainable forest management (e.g., The Montreal Process), the conservation of biological diversity (e.g., The Convention on Biological Diversity), and the protection of the environment (e.g., Agenda 21). Nationally, Carbon Monitoring System (CMS) data potentially contribute to Ministry for the Environment initiatives on Environmental Performance Indicators and State of Environment reporting, and to the biodiversity monitoring responsibilities of the Department of Conservation.

2.2 Survey design

The NZCMS uses a national grid-based network of permanent plots to provide an unbiased estimate of carbon stored in indigenous forests and in shrublands. Permanent plots offer significant advantages for monitoring change because they factor-out spatial variability that would otherwise mask temporal changes (Allen 1993; Coomes & Beets 1999).

In recognition of the value of existing long-term datasets, the NZCMS makes extensive use of existing permanent forest plots from the National Vegetation Survey (NVS) databank (Wiser et al. 2001), and uses NVS plot methodologies (Allen 1992, 1993) as the basis for its carbon and plant biodiversity monitoring protocols. NVS is a physical archive and computer databank managed by Landcare Research, Lincoln. It contains records from approximately 45 000 vegetation plots spanning more than 50 years of monitoring in New Zealand's terrestrial ecosystems. The geographic coverage of NVS is widespread but patchy. The databank provides extensive forest coverage for several major mountain ranges (e.g., Tararua, Ruahine, Kaimanawa) and partial coverage for others (e.g., Te Urewera, Fiordland), but some regions with substantial forested areas (e.g., SE Otago, NW Nelson, Northland) are significantly under-represented, as are lowland forests and shrubland communities (Wiser et al. 2001). In establishing the CMS plot network, existing NVS plots are used wherever they are available and new permanent plots established where they are not. All plots (new and existing) are measured using the protocols outlined in this manual.

2.2.1 Size of the plot network

The primary focus of the NZCMS is to monitor carbon sequestration rates (i.e., changes in carbon stocks per unit time), not to provide one-off estimates of carbon stocks. The number of plots required to estimate carbon sequestration rates within specified confidence limits cannot be determined until the plot network has been established and remeasured (Coomes et al. 2002). Based on data from remeasured NVS forest plots, Coomes & Beets (1999) estimate that for indigenous forests a national network of around 500 plots would be capable of detecting changes in carbon stocks of $<0.2\%$ per annum. The lack of an existing network of permanent plots precludes similar calculations being made for shrublands.

The initial estimate of national carbon stocks will use the 1996/97 version of the LandCover Database (LCDB1) to identify the areal extent of indigenous forest (6.25 million hectares) and shrubland (2.65 million hectares) vegetation in New Zealand. Superimposing an 8-km² grid on LCDB1 identifies c.1400 grid intersects within areas mapped as indigenous forest and shrubland. Because the purpose of the plot network is to provide an unbiased estimate of carbon stocks in areas mapped by LCDB as forest or shrubland, grid intersects outside these areas are not considered even if they contain woody vegetation. Based on calculations from a pilot trial that used a 9-km² grid (Coomes & Beets 1999), the 8-km² plot intensity is expected to provide a better than 95% probability that carbon stock estimates will be within 5% of the mean (± 10 tonnes/hectare) in indigenous forests, and 12% of the mean (± 2.6 t/ha) in shrublands.

Where existing NVS plots occur within 4 km of a grid intersect they are used as part of the CMS plot network. Where existing NVS plots are located along a transect, a plot from the transect nearest to the grid intersect is randomly chosen. For NVS plots that do not form part of a transect, the plot nearest to the grid intersect is chosen. NVS plots that double as the non-fenced control for an enclosure plot are excluded from the CMS plot network, on the grounds that these plots have traditionally been subjectively located. Where no NVS plot exists, or where an existing plot cannot be re-located, a new permanent plot is established at the grid intersect. Each plot has a unique identifier based on its position on the sampling grid. NVS plots are also identified by their existing NVS survey, line and plot number.

2.2.2 Sampling strategy

The initial measurement of NZCMS plots is to be completed over a period of 5 years. There are several ways in which this can be achieved. One option is to measure all CMS plots over 20% of the country each year. The downside to this approach is that whenever a national carbon estimate is calculated, data from some parts of the country will be less current than data from other areas. There are also potential problems associated with

factors (e.g., climate) that vary from year to year at regional or local scales. For these reasons we have chosen an alternative approach whereby a random set of CMS plots is selected (without replacement) for measurement each year. Using this sampling strategy the dataset for each or any combination of years will give a valid national running average that is unaffected by local or regional influences.

The scale of the sampling grid and the nature of much of the terrain is such that field teams will not be able to measure more than one plot in any one day. During the CMS pilot trial (Coomes & Beets 1999) field teams seldom travelled directly from one plot to the next or camped partway between plots. Rather they entered and exited the hill country at the most convenient location to the sample point, and then moved to the next point of entry. We are of the view that whether a team visits 100% of sites within 20% of the sample universe (first option), or 20% of sites within 100% of the sampling universe (second option) in a field season will have little effect on sampling efficiency.

2.3 Sample plot location

Before establishing or remeasuring a CMS plot ensure permission has been obtained from the owner of the land on which the plot location is situated, and from persons whose property must be crossed to reach the plot location. Record this information along with information on team personnel and time allocation on the CMS Metadata sheet (Appendix 1.).

It is important to establish the position of the plot as accurately as possible so future parties can re-locate it without difficulty. A Global Positioning System (GPS) unit will normally be used to establish the position of the plot. Because GPS units differ in their capabilities, and satellite availability also influences the accuracy of GPS-fixed positions, it is important to record:

- The make and model of the GPS unit used to fix the location of the plot.
- Whether the plot coordinates are a 2D or a 3D fix. 3D fixes are preferable and are normally more accurate.
- Whether the plot coordinates were obtained from a single-position fix (2D or 3D) or are the average of several position fixes, and the precision of the result. Averaged positions are more accurate than single-position fixes. The precision of a position fix is calculated from the geometric positions of the satellites used to calculate the fix. It does not include all sources of error and so is indicative only, but is nevertheless useful for re-location purposes.

When using a GPS unit to fix the position of a CMS plot, use the instrument to navigate to within approximately 30 m of the plot location. Then use a compass bearing and the horizontal distance reading displayed by the GPS unit to identify the exact location of the plot. This forms corner P of the 0.04-ha plot (see Fig. 1). Where a GPS unit is being used to determine direction, ensure the direction function is set to magnetic.

The above procedure is recommended because the accuracy with which a GPS unit can locate a specified point decreases as that point is approached. Once the position of the plot has been identified, use the GPS unit to refix the position or, where this is not possible, to refix the position of an identifiable point (e.g., a prominent landscape feature) at a specified distance and magnetic bearing from the plot. It is important that the refixed position is an averaged reading as this increases the accuracy for re-location purposes (Burrows 2000; <http://nvs.landcareresearch.co.nz/>).

2.3.1 Existing NVS plots

National Vegetation Survey plots are typically located at fixed intervals along permanently marked transects. The origin of each transect is selected on a restricted random basis so that the plots are representative of the area as a whole, and the transect runs from valley bottom to treeline (Allen 1993).

Where an existing NVS plot is to be remeasured as part of the CMS:

- Obtain a copy of the Recce description, stem diameter printout, map and aerial photo (where available) from previous remeasurements to help re-locate the transect and the plot. Note that plotsheets from the original measurement and from the first plot on the transect frequently contain the best information for locating the plot to be remeasured. This information is archived in the NVS databank at Landcare Research, Lincoln.
- Where necessary re-mark the origin of the transect, and replace the permolat flagging along the transect.
- Use a GPS unit to fix the position of the transect origin and the location of the plot.
- Update the identification information, including the approach notes and location diagram, on the CMS Recce plot sheet (Section 3.2.1; Appendix 2A).

Where an existing NVS plot cannot be re-located after 6 hours searching, a new CMS plot should be established at the appropriate grid intersect using the procedure described in Section 2.3.2.

2.3.2 New CMS plots

Where a new CMS plot is being established:

- Use a GPS unit to locate the position of the plot, and to identify landscape features (e.g., streams, spurs, bluffs) that will assist future field parties to re-locate the plot.
- Use permolat flagging to mark the route from a clearly identifiable landscape feature to the site of the plot. In difficult terrain this is often best done when leaving the plot.
- Mark the position of the plot on the appropriate topographic map (NZMS 260 series) and aerial photo (where available), and complete the identification information section on the Recce plotsheet (see Section 3.2.1; Appendix 2A).

Where the field team is unable to establish a plot at the specified grid coordinate because it is either unsafe to do so (e.g., steep terrain prevents access to all or part of the plot area) or the surrounding terrain means it is impossible to get to the exact location:

- Do not attempt to establish the plot at the specified grid coordinate.
- Record a brief description of the physical characteristics of the site and the presence of any woody vegetation on the Recce plot sheet (Appendix 2A & B). This enables the grid intersect to be classed for carbon accounting purposes as either a nil value (no woody vegetation present) or a missing value (woody vegetation present, but unable to be measured).
- Where similar vegetation occurs within 200 m of the specified grid coordinate, re-locate the plot taking into account the altitude, aspect and vegetative cover of the specified grid coordinate.
- Where similar vegetation does not occur within 200 m of the specified grid coordinate, abandon the grid intersect in question.

3 Plot Layout and Description

Field data required to estimate carbon stocks in indigenous forests and in shrublands are collected from permanent 0.04-ha (20 × 20 m) plots. Protocols for forest plot measurements are based on Allen (1993).

Because a 0.04-ha plot size may not be big enough to adequately sample large trees, trees greater than 60 cm dbh are sampled on a larger, circular (approx. 0.13-ha) plot centred on the 0.04-ha plot (Fig. 1). A list of equipment needed to lay out and measure these plots is provided in Appendix 3.

Field data from CMS plots are also used to estimate a range of biodiversity parameters (e.g., vegetation composition and structure, sapling and seedling density). Protocols for the collection of these data have been adapted from Allen (1992,1993).

Before commencing to measure a CMS plot, decide which of the methods outlined in this manual (e.g., stem diameter and height, coarse woody debris, discrete or continuous shrubland) are most appropriate for the type of vegetation on the plot. Always complete the Metadata (Appendix 1), Recce (Appendix 2A & B), Plot Layout (Appendix 4) and Understorey (Appendix 10) record sheets regardless of vegetation type. Appendix 3 includes a list of the record sheets to be used for each of the CMS methods. Complete all fields on each sheet. Record data as neatly as possible, to minimise the chances it will be misread by the data entry operator.

3.1 Plot layout

To establish a 0.04-ha plot:

- Using the grid intercept as corner P, identify the bearing that runs along the predominant contour of the slope. Lay a 20-m tape along this bearing to form the lower boundary of the plot (P–M in Fig.1). Where the terrain is flat (i.e., there is no discernible slope) orient the P–M boundary along the north-south axis. Lay two 20-m tapes at right angles to the first, one at each end. Join the open ends with a fourth 20-m tape to form a 20 × 20 m square. Tapes should be pulled tight when laying out a plot on even ground. When the plot is in a gully or on a ridge etc. the tapes should follow the ground surface. Use a 3-4-5 triangle to ensure the plot corners are at right angles.
- Measure and record on the Recce plot sheet (Appendix 2A) the bearing and horizontal distance of the four plot axes (A–D, D–M, M–P, P–A) for both new CMS and existing NVS plots. When using a hypsometer (e.g., vertex) to make these measurements be sure to check the calibration of the instrument¹. If a hypsometer is unavailable, measure

¹ Note that a vertex hypsometer should be calibrated only after the instrument has equilibrated to the outside temperature. Recalibration is also necessary after a marked change in humidity (e.g., rain following a dry spell of weather).

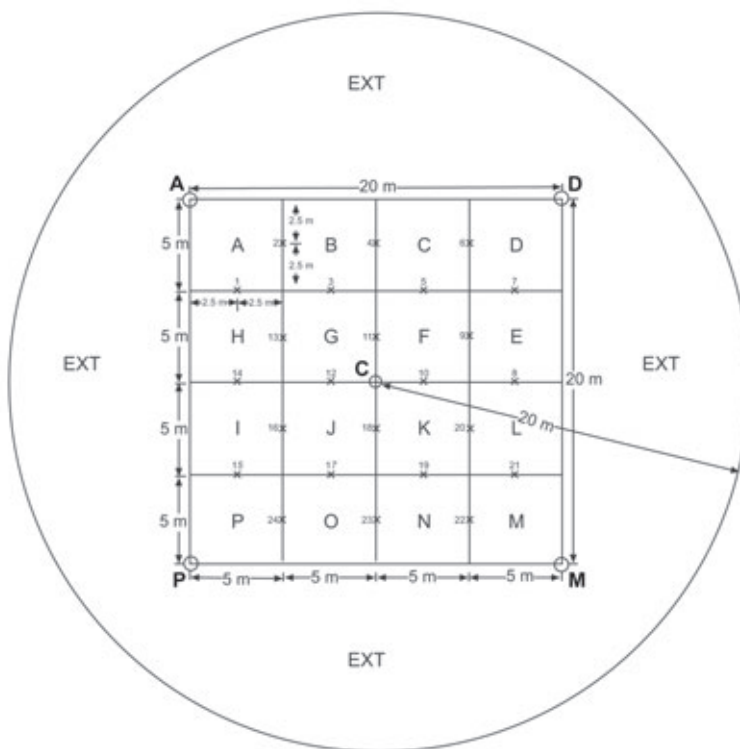


Figure 1. Layout of 0.04-ha (20×20 m) and 0.13-ha (20-m radius) plots (Redrawn from Allen 1993). See Sections 3.1 and 4.1.1 for an explanation of the symbols and the letter/number sequences.

and record the slope distance and slope angle for each axis. **Do not realign the axes of existing plots, because this will invalidate comparisons with earlier measurements.** If convex terrain obstructs the line of sight, place the hypsometer transponder on the obstructing highpoint and measure the horizontal distance of the axis in two stages. Where a plot axis is measured in more than one step, ensure that each segment has the same bearing.

- Permanently mark the centre (C) and the corners (A, D, M, P) of the plot with strips of permolat attached to large-sized aluminium pegs (see Appendix 3), and labelled with the appropriate letter (see Fig. 1). Bend the top of the aluminium pegs (this also

applies to the pegs used to mark seedling plots) to reduce the possibility that the permolat markers will fall off. Where plot or seedling pegs could pose a hazard to stock (e.g., where the plot includes pasture) drive them in to just below ground level.

- Nail a permolat strip to a tree (or other clearly visible object) near each corner peg, but outside the plot. Leave at least 2 cm of the nail protruding to allow for tree growth. On these permolat strips record the distance and magnetic bearing from the tree base to the nearest peg, and use an arrow to indicate the direction of the peg. Record this information on the Plot Layout sheet (Appendix 4).
- Subdivide the plot into 16 equal-sized (5 × 5 m) squares using additional tapes or nylon cords. Tapes should be pulled tight when laying out a plot on even ground. When the plot is in a gully or on a ridge etc. the tapes should follow the ground surface. Label the 16 subplots from A to P on the plot sheets starting in the top left-hand corner (Fig. 1).
- Record the dimensions of the plot on the CMS Plot Layout sheet (Appendix 4).

To establish a circular 0.13-ha plot:

- Use a hypsometer to identify all trees and logs >60 cm in diameter that are within a 20-m radius of the centre of the 0.04-ha plot. In the case of trees on the margins of the plot measure the horizontal distance to the mid-point of the tree to determine whether it is within the circular plot. The area outside the 0.04-ha plot but within the 0.13-ha plot is labelled EXT (Fig. 1).

Keep disturbance to the plot and its immediate surroundings to a minimum, to minimise the possibility that changes observed on future visits may result from previous measurement activities.

3.2 Plot description

This describes the physical and biological characteristics of the 0.04-ha plot. Protocols are adapted from the reconnaissance plot method described by Allen (1992). These data are collected on all CMS plots. Appendix 2A contains an example of a completed plot sheet.

3.2.1 Identification information

- CMS plot identifier: This is the unique letter/number code (e.g., BD112) that identifies the position of the plot on the CMS's 8-km² sampling grid.
- Catchment/Area/Region: Record the name of the catchment, area and region in which the plot is located, e.g., Chinkey Creek, Inangahua River, Westland. In the case of existing NVS plots include the name of the NVS survey dataset.

- **Topomap/GPS reference:** Record the NZMS 260 map sheet number and name (e.g., L30, Reefton), and GPS reference using the seven-figure NZ Map Grid coordinates (e.g., 2416048, 5901060). (See Section 2.3 for further information on the use of GPS units for plot location). Mark this point on the map and (where available) aerial photo.
- **Aerial photo:** Record the survey name, run and number of the aerial photo on which the plot location is marked (if available).
- **Landcover/land-use class:** Record the expected landcover/land-use class for the plot as shown on the version of the Land Cover Database (LCDB) being used for the current carbon estimate, and add (in brackets) your assessment of the actual landcover class. For example, a plot identified as forest on LCDB1 but found to be shrubland in the field would be recorded as Forest (Shrubland)². A list of the landcover/land-use classes used by LCDB1 is given in Appendix 5.

LCDB1 defines shrubland as woody vegetation in which the cover of shrubs and trees in the canopy is >20% and in which shrub cover exceeds that of any other growth form or bare ground. Shrubs are woody plants (e.g., kanuka, manuka, matagouri, gorse, broom, hawthorn) generally <10 cm dbh (diameter at breast height). Indigenous forest is defined by LCDB1 as natural forest cover dominated by indigenous tall forest canopy species. The minimum mapping unit for LCDB1 is one hectare, the target classification accuracy is >90%, and the positional accuracy is ± 25 metres.

- **Measured by:** Record the full name (e.g., James Barton) of the person(s) doing the description of the plot.
- **Recorded by:** Record the full name of the person recording the descriptive data.
- **Date:** Record day/month/year in full (e.g., 29 April 2002)
- **Approach:** Record a description of how to get to the site. Include information on the vegetation and physiography, and GPS grid references of prominent landscape features. The description should be sufficiently detailed to enable people who have not previously been to the plot to locate it without extensive searching.
- **Location diagram:** Sketch the route to the plot emphasising prominent landscape features (e.g., slips, gullies, bluffs, roads, streams, large treefall gaps). Where possible include GPS grid references for significant way-points. Centre the sketch on the plot location. Include transects, and distinguishing vegetation features. Wherever possible supplement this with a photographic record, to assist future parties to locate the plot.

² Note that the LCDB classification for a plot (forest, shrubland) may differ from the methods (forest, shrubland) used to measure the woody biomass on that plot (see Section 4).

- Notes: Provide a two-sentence description of the vegetation on the plot and record any additional observations (e.g., changes in physiography) and impressions (e.g., damage caused by wind, snow, erosion, or animals).

3.2.2 Site description

The first five parameters should be measured from the centre of the plot.

- Altitude: Measure to the nearest 10 m using an altimeter. Barometric altimeters should be corrected each morning, or more frequently during changeable weather, from points of known altitude on the topographical map. Do not use a GPS unit to measure altitude, or infer plot height from a topographical map.
- Physiography: Describe, using one of the following categories: Ridge (including spurs), Face, Gully or Terrace. If more than one category could apply, circle the predominant option. Where there is a major change in physiography within the plot, record this information (and the associated aspect measurements) in the Notes section.
- Aspect: Determine the physiography of the plot before measuring the aspect. Use a compass to measure the predominant aspect of the plot to the nearest 5 degrees. An aspect measurement is taken at right angles to the general lie of the plot. Where there is a major change in aspect across the plot, record the measurements (and the associated physiography information) in the Notes section. For plots with a slope of <5 degrees, aspect cannot be determined and should be recorded as X. **Do not use zero to indicate a missing value or the aspect of a flat site.**
- Slope: Measure the average slope of the plot along the predominant aspect to the nearest degree, using an Abney level or equivalent instrument. Sight the instrument on an object at eye level 10 m upslope and 10 m downslope (i.e. at the edge of the 0.04-ha plot), and average the readings. The shape of the slope, along the predominant aspect, should be described as Convex, Concave, or Linear.
- Horizon angles: Use an Abney level or equivalent instrument to measure the angle from the centre of the plot to the horizon at eight equidistant (45°) points. Record angles above the horizontal as plus (+), and those below the horizontal as minus (-). Where the horizon is obscured (e.g., by fog or low cloud), estimate the horizon angle and note that the recorded value is an estimate. When all eight measurements are summed, low values indicate exposed sites (e.g., ridges) and high values indicate protected sites (e.g., gullies) within the wider landscape (McNab 1993).
- Parent material: Record the types of bedrock you see on the plot (on site). Then use the Geological Map of New Zealand (1:250,000) and the rock type classification for the New Zealand Land Resource Inventory (Table 7 in Lynn & Crippen 1991) to identify

the underlying parent material (mapped). Note any disagreement with the broad map classification; this will be particularly important where there are extrusive/intrusive rocks.

- Soil depth: Measure the depth of the soil with a probe. Take five measurements, one at the centre of the plot (C) and one halfway between the centre and each of the four plot corners. For example, soil depth position 'A' is at the intersection of subplots A, B, H, and G (see Fig. 1). Insert the probe at right angles to the slope. If the probe hits tree roots move its position slightly. Record soil depth measurements to the nearest 5 cm, and any measurement over 50 cm as 50+. Label as A, D, M, P and C.
- Surface stability:
For the whole plot record:
 - Rock on surface – presence or absence of surface rock.
 - Bedrock on surface – determine whether the surface rock is exposed bedrock.
 - Percentage of bedrock, broken rock and soil at the ground surface. These values should sum to 100%.
 - Size of loose rock – size of broken rock (if any) on surface as greater and/or less than 30 cm diameter.
 - Mode of transport – classify broken rock (if possible) as mostly deposited as a result of alluvial (river or glacial deposits), colluvial (erosion debris), or volcanic activity.
- Drainage: Record as Good (fast runoff and little accumulation of water after rain), Moderate (runoff is slow, water accumulates in hollows for several days), or Poor (water stands for lengthy periods).

3.2.3 Vegetation parameters

- Ground cover: Estimate the percentage (to the nearest 5%) of the plot below 1.35 m that is covered by:
 - Vegetation live vascular vegetation including foliage, tree trunks and exposed roots
 - Moss non-vascular vegetation
 - Litter litter including logs and branches
 - Bare ground exposed soil (not covered by litter, vegetation, moss, or rocks)
 - Rock exposed rock on surface, either broken rock or bedrock that is not covered by vegetation, moss or litter.

NB: Tree trunks and exposed roots form a very small fraction (usually <1%) of the vegetative cover (see Appendix 6).

Because ground cover includes all vegetation below 1.35 m, estimates may sum to >100% where there is more than one layer of vegetative cover. Where necessary make notes to clarify the situation.

- Mean top height: Measure the average height of the dominant canopy trees to the nearest metre using a rangefinder.
- Canopy percentage: Use the Foliage Cover Scale (Payton et al. 1999; Appendix 6) to estimate to the nearest 10% the percentage of sky blocked out by the vegetation. Take an estimate at each of the five points used to measure soil depth on the 0.04-ha plot, and record the average. Each estimate should be taken from a height of 1.35 m (i.e., for vegetation less than 1.35 m record as 0).

3.2.4 Additional biological information

- Cultural: Record direct evidence (e.g., cut stumps, mine tailings) of human interference using one or more of the following categories – None, Fire, Logged, Cleared, Mined, Grazed, Tracked. Use the Notes section to justify your choice(s), and record indirect evidence (e.g., plant species characteristic of post-fire communities as evidence of a past fire) of human activity.
- Browse: Record conspicuous browsing damage caused by vertebrate and invertebrate herbivores to plant species on the plot (Appendix 2B) using the following categories:
 - Light (L) less than 10% of leaves browsed on all plants of the species
 - Moderate (M) 10–50% of leaves browsed on all plants of the species
 - Heavy (H) more than 50% leaves browsed on all plants of the species.

Record the herbivore responsible where this can be reliably determined, or state unknown. A description of browse damage (e.g., extent of bark stripping and height of browsing) should be made in the Notes section.

- Fauna: Record the presence of mammal/bird/reptile/invertebrate species that can be positively identified by sight or sound (Appendix 2B).

3.3 Vegetation description

Plant species composition is described in six height tiers, with a seventh tier reserved for epiphytes. Protocols largely follow the Recce plot method described by Allen (1992), except that the description is confined to the 0.04-ha area of the CMS plot. These data are collected on all CMS plots. To ease the task of the data entry operator, record data as neatly as possible.

Plant species names are recorded using a code made up from the first three letters of the plant genus (upper case) followed by the first three letters of the species, subspecies or variety (lower case). For example, *Metrosideros umbellata* (southern rata) is written as METumb, and *Nothofagus solandri* var. *cliffortioides* is written as NOTcli (see Hall 1992). Where only the genus is known (e.g., *Metrosideros*), use the first six letters written in upper case (e.g., METROS). Use only recognised published taxa. Plant names in the New Zealand flora are available from the New Zealand Plant Names database on the Landcare Research website (<http://nzflora.landcareresearch.co.nz/>). In some cases the intuitive six-letter code applies to more than one taxon (e.g., *Pseudopanax colensoi* and *Pseudowintera colorata*). Where there is a possibility of ambiguity either use the non-standard six-letter code listed in Appendix 7 (vascular plants only), or write the plant name in full.

Where a species is unknown, collect a specimen with as many vegetative and floral parts as practical. Record the species with a tag name that reflects a feature of the plant. Tag names are specific to the plot at which they are created, and should be used consistently throughout the records (e.g., recce, stem diameter, seedling) for that plot. Use © to indicate the presence of a specimen requiring identification wherever this species occurs on the datasheets. Where possible collect specimens from outside the plot. Do not collect plant specimens from understorey subplots. Label all plant specimens with their tag name, plot number and collection date, and place vascular plants between sheets of absorbent paper in a field press. The use of a plant press in the field reduces the likelihood of misplaced specimens, and produces higher-quality samples for later identification. Non-vascular plants should be stored in paper bags with labels that include substrate information.

On the reverse side of the CMS Reconnaissance (Recce) plot sheet (Appendix 2B), record the plant species composition of the plot by height tier. Plant species are deemed to be present in a tier when they have foliage present within that tier. This includes plants rooted outside the plot but with foliage overhanging the plot. To avoid repetitive recording, use one row per species. Where the number of plant species exceeds the number of rows on the datasheet, use a second data sheet to complete the plot description. Ensure both datasheets contain the same header information (e.g., plot identifier, date), and that they are cross referenced (e.g., page 1 of 2).

Take care to ensure that plant species are recorded in the correct tier. This is especially important where data are to be used to identify past and present impacts of mammalian herbivores such as deer, goats or possums.

- Tier One (T1): Record all canopy trees taller than 25 m in this tier. Measure height range in metres and diameter range at breast height in centimetres (e.g., 28–30 m high, 60–100 cm dbh). Use the diameter and height data collected on the plot, to derive this information. Breast height is defined as 1.35 m above ground level, and is measured on the uphill side of the tree. Estimate and record the total foliage cover in the horizontal plane for all species in the tier using the Braun-Blanquet cover-abundance scale outlined in Table 1. Assign a cover class to each species listed in the tier.
- Tier Two (T2): Record all vascular plant species between 12 and 25 m tall. Measure and record height and diameter ranges as for T1. Estimate the total cover of all species in the tier, and assign a cover class to each species listed in the tier.
- Tier Three (T3): Record all vascular plant species 5–12 m tall. In a forest this tier would include understorey small tree species, large shrubs, and tree ferns. When the canopy occurs in this tier record as for T2. Estimate the total cover of all species in the tier, and assign a cover class to each species listed in the tier.
- Tier Four (T4): Record all vascular plant species between 2 and 5 m tall. Estimate the total cover of all species in the tier, and assign a cover class to each species listed in the tier.
- Tier Five (T5): Record all vascular plant species between 30 cm and 2 m tall. Estimate the total cover of all species in the tier, and assign a cover class to each species listed in the tier.
- Tier Six (T6): Record all vascular plant species up to 30 cm tall in the ‘vascular’ column. Estimate the total cover of all species in the tier, and assign a cover class to each species listed in the tier. Record the moss, liverwort, and lichen species that provide the majority of the non-vascular plant cover on the plot in the ‘non-vascular’ column. Distinguish between non-vascular species growing on the ground (terrestrial), and those growing on ground-based substrates (terrestrial on woody debris). Do not assign a cover class to non-vascular plant species.

Visually estimate percentage foliage and record in one of six classes. Do not include tree trunks in cover class estimates. To maximise the consistency of cover class data, cover estimates should be checked periodically against a Foliage Cover Scale (Appendix 6).

- Tier Seven (T7): Record all species of vascular epiphytes (plants growing on other living plants) within the plot in the ‘vascular’ column. Record the moss, liverwort, and lichen species that provide the majority of the non-vascular plant cover on stems,

Table 1. Cover classes modified from the Braun-Blanquet cover-abundance scale (see Mueller-Dombois & Ellenberg 1974).

Cover Class	Percentage Cover	Equivalent Surface Area (20 × 20 m plot)
1	<1	< 2 × 2 m
2	1–5	< 1 (5 × 5 m) subplot
3	6–25	1–4 subplots
4	26–50	4–8 subplots ($\frac{1}{4}$ – $\frac{1}{2}$ plot)
5	51–75	8–12 subplots ($\frac{1}{2}$ – $\frac{3}{4}$ plot)
6	76–100	12–16 subplots (> $\frac{3}{4}$ plot)

twigs and leaves up to a height of 2 metres. Do not estimate cover for vascular or non-vascular plant species in this tier.

- Lianes: Record these in the tier(s) in which they occur. For example, if the majority of the foliage on a *Clematis vitalba* (old man's beard) vine is smothering the canopy of a 10-m-high tree, the presence and cover class of that species will be recorded in T3 (5–12 m).

4 Vegetation Measurement, Sample Collection, Storage and Processing

The methods used for measuring CMS plots are dictated by the structure of the vegetation (shrub vs tree), not by whether the plot is identified as forest or shrubland in the current LCDB classification. They do not include methods for below-ground biomass/carbon estimation, which are dealt with separately as part of the analysis procedure (Coomes et al. 2002).

4.1 Trees, tree ferns and lianes

Diameter (dbh) and height data (Appendix 8) are used to calculate carbon estimates for live and dead standing trees, tree ferns and lianes. Data collection protocols are based on Allen (1993). A tree or tree fern is defined as being within the plot if at least 50% of its trunk is within the bounded plot area. Lianes are recorded where they are rooted within the plot.

Where there are a large number of stems on a plot, use chalk or some other non-permanent marker to keep track of which stems have been tagged and measured. This reduces the likelihood that stems will be missed or, in the case of saplings, double counted.

Where data are recorded by subplot, leave a blank line between data from successive subplots. This minimises the chances of data being assigned to wrong subplot, and eases the task of the data entry operator.

4.1.1 Diameter measurements

What to tag and measure

Stems

- Tag and measure the stem diameter at breast height (1.35 m) of all live trees (≥ 2.5 cm dbh), dead standing trees (≥ 10 cm dbh), and tree ferns (with stems over 1.35 m tall) within the 0.04-ha (20 × 20 m) plot. Stems less than 2.5 cm dbh are classed as saplings (see Section 4.3). Note that on sloping plots breast height is measured from the uphill side of the tree.
- Tag and measure all trees (live and dead standing) greater than 60 cm dbh within the larger 0.13-ha (20-m radius) plot (see Fig. 2). Note that trees > 60 cm dbh are tagged and measured within a 20-m **horizontal** distance of the plot centre.
- Multi-leader stems and multiple stems within a clump. Treat each stem as an individual tree (Fig. 2c), and bracket stems belonging to the same tree (see example in Appendix 8). Unless stems are malformed or crowning occurs below breast height (see below), all stems should be tagged at the same height.

- Where a stem is leaning, it should be tagged and measured 1.35 m from the base, rather than at 1.35 m above ground level. Only measure leaning trees that are rooted within the plot.
- Where a live tree has fallen and resprouted, tag and measure the fallen stem as if it were an item of Coarse Woody Debris (see Section 4.4). Allocate 'living logs' a decay class of 0. Sprouts arising from 'living logs' should be classed as saplings (i.e., counted but not tagged or measured) where they are >1.35 m tall and <2.5 cm diameter at this height, and as stems (i.e., tagged and measured) where their diameter 1.35 m from the surface of the parent log is ≥ 2.5 cm.
- Living trees that have fallen into the plot should also be treated as 'living logs'.

Epicormic shoots, epiphytes and lianes

Epicormic shoots (also known as coppice shoots or water sprouts) are straight, unbranched stems originating from dormant buds under the bark, rather than from terminal or axillary buds on twigs. **Epiphytes** are plants growing on other plants. **Lianes** are woody, climbing plants.

- Epicormic shoots and epiphytes originating ≤ 1.35 m from the base of the stem are considered to originate below breast height, and those >1.35 m from the base of the stem are considered to originate above breast height.
- Epicormic shoots originating below breast height (1.35 m) are treated as saplings (i.e., counted but not tagged or measured) when the length of the shoot is >1.35 m and the diameter of the shoot at breast height (1.35 m) is <2.5 cm. They are treated as trees (i.e., tagged and measured) when their diameter at breast height is ≥ 2.5 cm. Record the presence of epicormic shoots in the Notes column of the Stem Diameter/Height record sheet (Appendix 8).
- Epicormic shoots originating above breast height are treated as branches and are not counted, tagged or measured.
- Epiphytes rooted below breast height are treated in the same manner as epicormic shoots. Record the presence of epiphytes in the Notes column of the Stem Diameter/Height record sheet (Appendix 8).
- Epiphytes rooted above breast height are recorded as part of the Recce description (Section 3.3), but are not tagged or measured.
- Lianes with stems >5 mm and <2.5 cm at breast height are treated as saplings (i.e., counted), and those ≥ 2.5 cm dbh as trees (i.e., tagged and measured). The 'breast height' of a liane is taken as 1.35 m from the base of the stem.

- Where lianes such as supplejack (*Ripogonum scandens*) or kiekie (*Freycinetia banksii*) form impenetrable thickets, they are treated as continuous shrubs for carbon accounting purposes (Sections 4.2.2 and 4.2.3) and recorded on the Recce description (Section 3.3), but are not tagged or measured.

Other circumstances when stem diameter is not measured at breast height

- Malformed stem. Measure the diameter as close to breast height as possible, alternating above (even-numbered tags) and below (odd-numbered tags) 1.35 m, where the stem becomes more regular (Fig. 2a).
- Crowning occurs below breast height. Measure stem diameter alternately above (even-numbered tags) and below (odd-numbered tags) the level of branching or nodal swelling, where the stem reverts to a more regular diameter (Fig. 2b). Where a stem that is to be tagged above breast height is <2.5 cm in diameter, tag and measure the stem and record the reason for the <2.5-cm measurement in the notes column.
- Whenever a stem is tagged at other than breast height, note the reason for the departure from the standard practice in the Notes column of the Stem Diameter/Height record sheet (Appendix 8).

How to tag and measure it

- Use a galvanised flathead nail to attach a numbered aluminium tag to each tree or tree fern stem at breast height (1.35 m). The nail should protrude at least 2 cm (to allow for growth), and be positioned so as to allow the tag to sit flush against the trunk. For tree ferns attach the tag using a long (>75 mm) nail or a loop of thin wire twisted around the trunk. Do not place the tag in a position where it is likely to be overgrown, e.g., between two leaders that are likely to fuse as the tree grows. On sloping plots, breast height is measured from the uphill side of the tree.
- Before measuring the stem diameter, clear moss and other debris in the vicinity of the tag, and loosen any lianes that will interfere with the diameter measurement. Take care not to damage the bark.
- Measure the diameter of the stem 1 cm above the tag. Keep the tape at right angles to the axis of the stem, as any deviation from this position will produce a positive bias in the measurement.
- Record the subplot (A–P), species code, tag number, and diameter for all trees/tree ferns (including dead standing stems) on the 0.04-ha plot, and for large (>60 cm dbh) trees on the 0.13-ha plot. Assign large trees that are outside the 0.04-ha plot to subplot EXT (Fig. 1). Ensure that the full tag sequence (e.g. AB02163) is recorded for

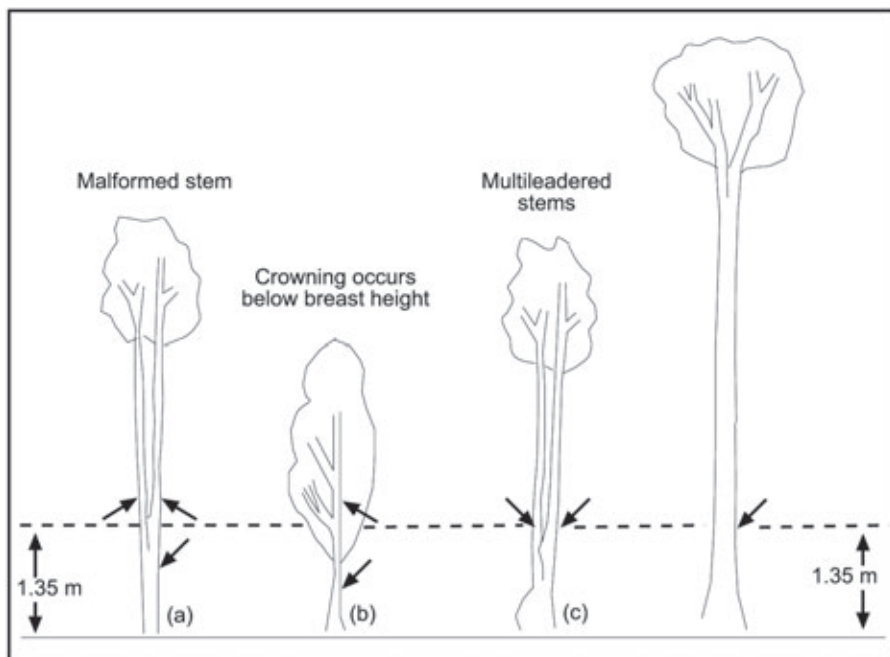


Figure 2. Diameter measurements: where to tag a tree (Modified from Allen 1993).

each individual to reduce the possibility of problems resulting from the use of different tag sequences on future remeasurements. Appendix 8 contains an example of a completed Stem Diameter/Height record sheet.

- Where a tree has died since the plot was last measured and is still standing, measure and record the diameter at breast height (on the Stem Diameter/Height record sheet) to enable the growth since the last measurement to be calculated. Write 'dead' in the Notes column, and where possible record the cause of death (e.g., windthrow, snapped, felled). Stems ≥ 10 cm dbh are classified as coarse woody debris (see Section 4.4). Do not remove or fold over tags on dead trees ≥ 10 cm dbh, as these stems will continue to be monitored as part of the Coarse Woody Debris pool.
- The use of different tag series for live standing stems and items of coarse woody debris is recommended as a means of more easily re-locating individuals at the next remeasurement.

4.1.2 Height measurements

- On each plot where they are present, measure the height (and diameter) of 15 individuals from each of the following groups: broadleaved trees, conifers, tree ferns, and dead standing stems. The sample should include the full diameter range for each group, and include malformed stems. Start by selecting one individual of each group from each of the 16 subplots in the 0.04-ha plot, and then choose further individuals to fill the gaps in the diameter range. This sample should include all trees ≥ 60 cm dbh or, where trees of this size are absent, the largest five trees on the plot.
- Measure tree height to the top of the crown, and tree fern height to the base of the live fronds. Measurements should be made to the nearest decimetre.
- For trees/tree ferns less than 10 m tall, height poles can be used. For taller trees Vertex or Suunto hypsometers are the recommended instruments for height measurements.
- To accurately measure the height of a tree it is necessary to be able to see the highest point of the tree crown, which may require moving some distance from the base of the trunk. This is especially true for trees with irregular or large rounded crowns. On sloping ground height measurements are usually best taken from a position upslope of the tree (Fig. 3).
- Wherever possible avoid leaning trees because their measurement can incur large errors. Where the measurement of leaning trees is unavoidable, measure their height at right angles to their lean. Where this isn't possible (e.g., a large rounded crown that can only be viewed from an upslope position) measure the height of the tree from the best vantage point.
- When measuring tree height, estimate the angle of lean to the nearest 10° for trees with a lean $>20^\circ$. Record the angle as minus (–) where the tree is leaning away from the observer, plus (+) where it is leaning towards the observer, and across (a) where the angle of lean is at right angles to the observer (see Appendix 8).
- Record the species, tag number, diameter, height and where appropriate the angle of lean for each individual on the CMS Stem Diameter/Height record sheet (Appendix 8).

4.2 Shrubs

For the purposes of CMS field sampling, shrubs are defined as woody plants that form part of the vegetation canopy, are greater than 30 cm in height, and have an architecture unsuited to stem diameter measurements. Shrubs may occur as isolated plants or discrete clumps (Section 4.2.1), continuous cover of one or more woody species (Section 4.2.2),

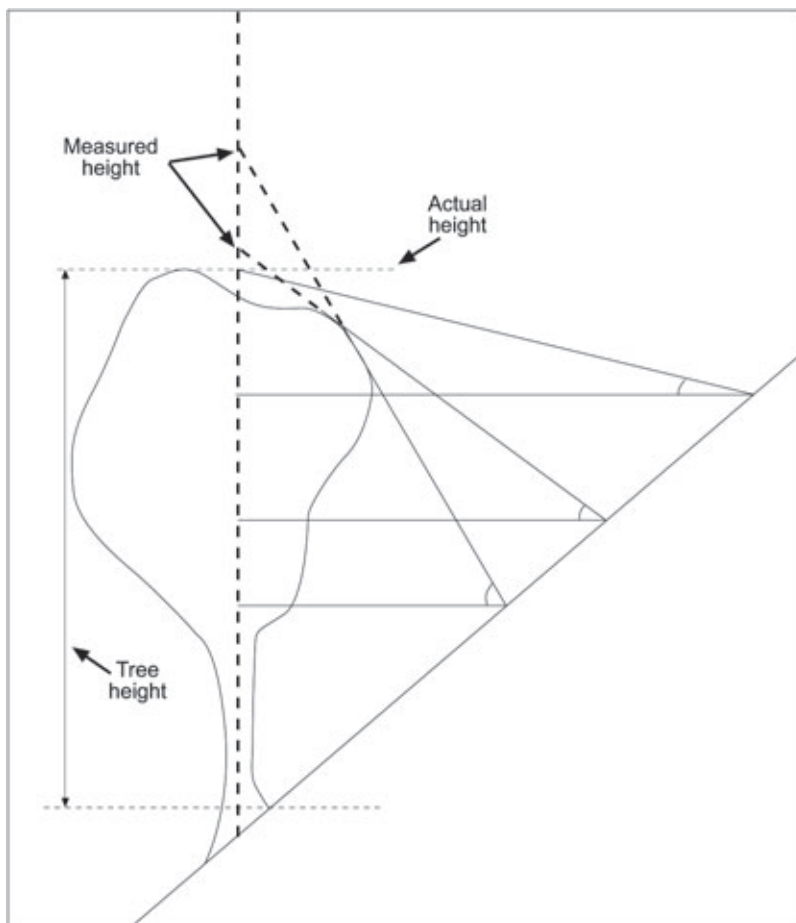


Figure 3. Height measurements: why it is important to be able to see the top of the tree.

or mixtures of discrete shrubs and continuous shrub cover (Section 4.2.3). Where shrublands originate after disturbance (e.g., fire, logging, erosion) in what was formerly a forested landscape they may include elements of the former vegetation (e.g., mature trees, logs) as well as those of the regenerating forest (e.g., saplings, small trees) (Section 4.2.4).

Shrubs growing under existing forest canopies form a minor part of the carbon pool and are not measured for carbon accounting purposes. Where shrubs, as defined above, form the canopy vegetation of all or part of the plot (i.e., contribute a significant proportion of the carbon present in the woody vegetation) they are measured using the protocols described in Sections 4.2.1 to 4.2.4.

The CMS uses two methods for estimating biomass/carbon in shrubland vegetation. Both use the standard 0.04-ha plot (Fig. 1). Ideally, only one of the methods described below should be used on any one plot. Where either method could be used, the discrete shrub method (Section 4.2.1) is the preferred option. Where both methods are used on the same plot, use only one method for any one 5 × 5-m subplot.

For all shrubland plots

- Complete the Metadata (Appendix 1), Recce (Appendix 2), Plot Layout (Appendix 4), Shrubland Plot Layout (Appendix 9) and Understorey (Appendix 10A & B) record sheets.
- Tag and measure (diameter and height) all live (stems ≥ 2.5 cm dbh) and dead standing (stems ≥ 10 cm dbh) trees that are present on the plot, including live and dead standing trees > 60 cm dbh that are within 20-m (horizontal distance) of the plot centre. Record these data on the Stem Diameter/Height record sheet (Appendix 8).
- Measure and where appropriate tag all coarse woody debris (logs and fallen branches ≥ 10 cm in diameter) (Section 4.4) that are present on the plot, including logs > 60 cm in diameter that are within 20-m (horizontal distance) of the plot centre. Record these data on the Coarse Woody Debris record sheet (Appendix 11).
- Decide whether the remaining woody vegetation on the plot is best measured for carbon accounting purposes as saplings (Section 4.1.1), discrete shrubs (Section 4.2.1), or continuous shrub cover (Section 4.2.2), and measure according to the relevant protocol.

4.2.1 Biomass assessment of discrete shrubs

Use this method in shrublands where individual shrubs or shrub-clumps (discrete clumps of one or more woody species) are clearly separable. An individual shrub is defined as being within the plot if at least 50% of its crown is within the bounded plot area.

Within the 0.04-ha plot:

- Measure the maximum vertical height and two orthogonal widths in the horizontal plane (one of which is the maximum width) of each individual shrub and shrub-clump (> 30 cm height) within the plot. Note that shrub height is the natural maximum height of the tallest stem.

- Record the subplot (A–P), species code, height, and two orthogonal widths for all shrubs/shrub-clumps as shown in Appendix 12.
- Where a shrub-clump consists of more than one species, list all species in order of their contribution to the total shrub volume.

Outside the 0.04-ha plot:

- Select four representative shrubs (or shrub-clumps) that are at least 3 m outside the 0.04-ha plot boundary for biomass determination. To ensure a representative selection identify an area of similar vegetative cover and site characteristics in the vicinity of the plot, and choose individual shrubs or shrub-clumps similar in height and width to those within the plot. Select shrub species approximately in proportion to their contribution to the shrub cover on the 0.04-ha plot. For example, if shrub species 1–5 comprise 50, 20, 20, 5, and 5% respectively of the cover on the 0.04-ha plot, select and label two shrubs of species 1, one shrub of species 2, and one shrub of species 3. Where rare plant species are encountered, use a common species with a similar habit for the biomass harvest.
- For each shrub record the plot identifier and date, shrub number (1–4), plant species (Appendix 13A), and location relative to the 0.04-ha plot (Appendix 9).
- Before harvesting measure and record the height and two orthogonal widths of each shrub/shrub-clump to be harvested, as described above.
- Harvest the entire above-ground biomass of each shrub/shrub-clump by cutting at ground level. Weigh the entire shrub using a spring balance and record the total fresh weight on the biomass harvest form (Appendix 13A). Shrubs with a fresh weight >5 kg should be weighed to the nearest 100 g. Whole plants or subsamples (see below) weighing <5 kg should be weighed to the nearest 25 g.
- Wherever possible the entire harvested shrub should be returned to the laboratory for dry weight and carbon determination. Where this creates logistical difficulties, shrubs with a fresh weight >1 kg may be subsampled as follows:
 - Divide the harvested shrub into separate piles of uniform plant material (e.g., leafy shoots, bare branches, stems) on a tarpaulin. If the woody material is not particularly uniform in length or diameter, cut into uniform lengths and order from smallest to largest in diameter.
 - Take a subsample (e.g., $\frac{1}{4}$ or $\frac{1}{2}$) by randomly keeping every second or fourth wood section. This method of subsampling can also be used for leafy shoots. Record the subsampling procedure on the Biomass Harvest record sheet (Appendix 13A) as a check on the total and subsample fresh weight of the shrub.

- If equal-sized subsamples (e.g., $\frac{1}{4}$) are taken from each fraction (e.g., leafy shoots, branches, stems), then the fractions can be combined and treated as a subsample of the whole plant.
- Immediately weigh the subsample (0.5–1.0 kg) of each component in the field and record the weights on the Biomass Harvest record sheet. Place each subsample in a separate, sealed polythene bag, label, and return to the laboratory for oven-drying to constant weight in a forced-draft oven (65–75°C, but not >80°C). Where samples are not immediately returned to the laboratory, they should be held in cool storage (periods ≤ 7 days) or frozen (periods >7 days).
- Note that where whole shrubs or subsamples are weighed while in a bag or tarpaulin, the net weight should be recorded on the Biomass Harvest record sheet.
- Repeat the procedure for each harvested shrub. Ensure that the dimensions of the shrub are measured and recorded before harvesting, and that bags containing biomass samples are correctly labelled. Potential sources of error are more readily resolved if shrub samples are numbered, weighed, recorded and bagged separately.
- Record the data as shown in Appendix 13A.

4.2.2 Biomass assessment of continuous shrub cover

Use this method in shrublands where it is not possible to distinguish individual shrubs or shrub-clumps. Where shrubland is very dense, plot tapes may be strung over the shrub canopy, and pegs placed vertically below the subplot corners. In continuous shrubland, shrubs are defined as being within the plot if they occur within the conceptual rectangular box formed by the tapes that define the plot boundaries. This is in contrast to the individual shrub methodology, which defines a shrub as being within the plot if at least 50% of the crown is within the bounded plot area (Section 4.2.1).

Within the 0.04-ha plot:

- Measure the cover of woody vegetation on each of the 16 subplots of the 0.04-ha plot (Fig. 4a, Appendix 9). If the subplot is entirely covered with woody vegetation greater than 30 cm in height, cross-hatch the subplot and record 5 × 5 m under the subplot letter code. For subplots that are only partly covered by woody vegetation, cross-hatch the portion of the subplot covered by woody vegetation greater than 30 cm in height. Record measurements of the cross-hatched areas under the appropriate subplot label. Keep the measurements simple (e.g., rectangles, squares, right-angled triangles). Appendix 9 contains an example of a completed Shrubland Plot Layout sheet.

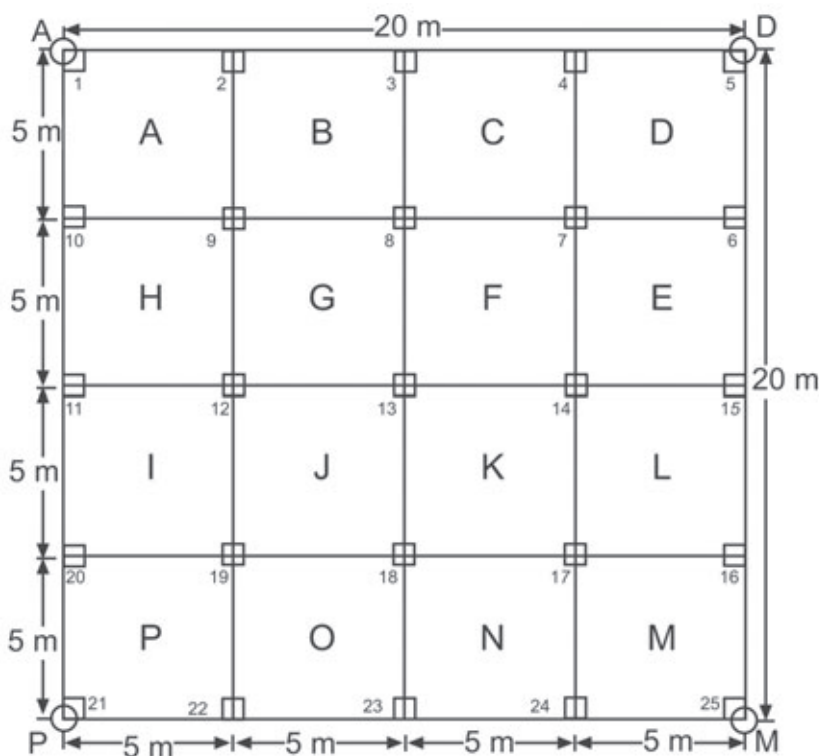


Figure 4a. Layout of height/cover subplots within a 0.04-ha plot.

- Within a 0.2×0.2 -m quadrat placed successively over each of the 25 grid intersects on the 0.04-ha plot (Fig. 4a), measure the maximum vertical height of the woody vegetation (to the nearest 10 cm) using a graduated pole, estimate the foliage cover (to the nearest 10%) using the Foliage Cover Scale in Appendix 6 (Payton et al. 1999), and identify the woody shrub species present. Where a quadrat contains no woody vegetation, both height and cover are recorded as zero. Except in low-statured vegetation, foliage cover should be assessed from the ground looking skywards. If the woody vegetation is less than 30 cm in height, then height and cover are recorded as zero. Record the quadrat number (1–25), species code(s), maximum shrub height and Foliage Cover value for each 0.2×0.2 -m quadrat on the Continuous Shrub Cover record sheet (Appendix 14).

Outside the 0.04-ha plot:

- Lay out a 4×1 -m biomass harvest plot in an area of continuous shrubland that is representative of the vegetation composition and site characteristics of the 0.04-ha plot, and that is at least 3 m away from the plot boundary.
- On gentle to moderately sloping ground measure the area of the biomass plot on a horizontal basis. Where steeper slopes make this impractical, lay the tapes parallel to the ground surface and measure the slope distance and slope angle for each axis. Record the method used to determine the area of the biomass harvest plot (tape distance or horizontal distance) and where appropriate the data on tape distance and slope angle in the plot dimensions section of the Continuous Shrub Cover record sheet (Appendix 14).
- First measure and then harvest all the woody vegetation within the 4×1 -m plot following the procedures described below.
- Within a 0.2×0.2 -m quadrat placed successively over each of the 10 grid intersects on the 4×1 -m biomass harvest plot (Fig. 4b) measure the maximum vertical height of the woody vegetation (to the nearest 10 cm) using a graduated pole, estimate the foliage cover (to the nearest 10%) using the Foliage Cover Scale in Appendix 6 (Payton et al. 1999), and identify the woody shrub species present. Except in low-statured vegetation, foliage cover should be assessed from the ground looking skywards. If the woody vegetation is less than 30 cm in height, then height and cover are recorded as zero. Record the quadrat number (1–10), species code(s), maximum shrub height and Foliage Cover value for each 0.2×0.2 -m quadrat. Appendix 14 contains an example of a completed plot sheet.

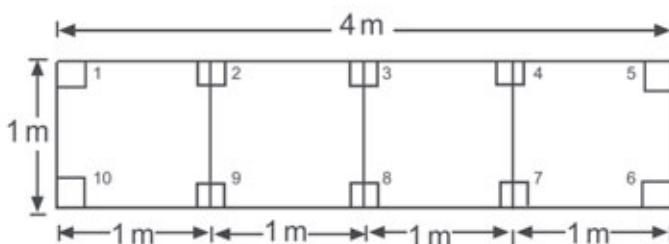


Figure 4b. Layout of height/cover subplots within a 4×1 -m biomass harvest plot.

- Harvest the entire above-ground biomass of the 4×1 -m plot, taking care to include only those parts of the aerial biomass that fall within the vertical rectangular box defined by the biomass plot boundary (i.e., in some cases only part of a shrub will be harvested). In other words, use the same definition of shrub biomass for both the 4×1 -m biomass plot and the 20×20 -m shrub plot.

- Measure the fresh weight of the biomass sample and take representative component subsamples for moisture content determination using the methods described in Section 4.2.1. Record the data as shown in Appendix 13B.

4.2.3 Biomass assessment of discrete and continuous shrub cover on the same plot

Use this method where areas of both discrete and continuous shrubland are present within the 0.04-ha plot.

Within the 0.04-ha plot:

- In any one subplot woody vegetation should be measured as either discrete shrubs or continuous shrub cover.
- For subplots (A–P) that contain discrete shrubs, measure the maximum vertical height and two orthogonal widths in the horizontal plane (one of which is the maximum width) of each shrub and shrub-clump, and record these data on the Discrete Shrub record sheet (Appendix 12).
- For subplots (A–P) that contain continuous shrub cover, measure the cover of woody vegetation as described in Section 4.2.2, and record these data on the Shrubland Plot Layout sheet (Appendix 9).
- Measure the maximum vertical height of the woody vegetation and assess the foliage cover of a 0.2 × 0.2-m quadrat at each of 25 grid intersects within the 0.04-ha plot (Fig. 4a, Appendix 9) as described in Section 4.2.2. Record these data on the Continuous Shrub Cover record sheet (Appendix 14). Where there is no shrub cover at a grid intersect or the woody vegetation is <30 cm high, record zero for both cover and height.

Outside the 0.04-ha plot:

- Use the biomass harvest method (discrete or continuous shrub cover) that represents the majority of the shrubland vegetation on the plot.

4.2.4 Biomass assessment of regenerating forest

Use this method where plant communities exhibit the characteristics of both forest and shrubland vegetation.

- Tag and measure (diameter and height) any trees (stems ≥ 2.5 cm dbh) that are present on the plot, including trees >60 cm dbh that are within 20 m (horizontal distance) of the plot centre. Record these data on the Stem Diameter/Height record sheet (Appendix 8).
- Measure (diameter and height/length) and where appropriate tag all standing dead trees and logs (≥ 10 cm in diameter) that are present on the plot, including logs >60 cm in diameter that are within 20 m (horizontal distance) of the plot centre. Record

standing dead trees on the Stem Diameter/Height record sheet, and logs on the Coarse Woody Debris record sheet (Appendices 8 & 11).

- Excluding the tagged and measured tree stems, decide whether the remaining woody vegetation on the plot is best measured for carbon accounting purposes as discrete shrubs (Section 4.2.1), continuous shrub cover (Section 4.2.2), mixed discrete and continuous shrub cover (Section 4.2.3) or forest understorey vegetation (Section 4.3.2), and proceed accordingly.

4.3 Saplings, seedlings and other understorey vegetation

These data contribute to the biodiversity monitoring objectives of the CMS, but are not used to estimate carbon stocks. Protocols have been adapted from Allen (1993). Data collection is restricted to the 0.04-ha plot.

4.3.1 Seedlings

Seedling and other understorey data are collected from all CMS plots.

- Herbs, ferns, seedlings of woody species and non-vascular plants are recorded on 24 (0.75-m²) circular understorey or seedling subplots (radius 49 cm) located midway between the intersection points of the tapes used to divide the 0.04-ha plot into 16 subplots (Fig. 1). Data are recorded on the Understorey Record Sheet (Appendix 10A & B).
- Permanently mark the centre of each subplot with a small-sized aluminium peg (see Appendix 3) and a permolat strip labelled with the subplot number. Bend the top of the aluminium pegs to reduce the possibility that the permolat marker will fall off.
- Where it is impossible to insert a subplot peg at the specified location (e.g., where the point falls on a tree root or a solid log), shift the subplot to the nearest point along the tape that the peg can be inserted.
- Where more than one substrate is available at the specified location (e.g., a suspended log), locate the subplot on the lowest substrate.
- Record the position of all subplots on the CMS Plot Layout sheet (Appendix 4). Where existing NVS plots use a different numbering system, retain the existing system and note this on the Plot Layout sheet (see Section 5.2).
- Use a piece of string to describe the area of the subplot. The understorey subplot radius (49 cm) is measured from the base of the seedling peg, with the string pulled tight but following the contour of the ground surface. Make a loop at one end of the string and place this over the seedling peg. Tie a knot 49 cm along the string to

denote the edge of the subplot. Check the length of the string before each fieldtrip.

- Record woody species less than 15 cm high by presence (✓) only. For woody species greater than 15 cm high, count and record the number of stems within each of the following height classes: 16–45 cm, 46–75 cm, 76–105 cm and 106–135 cm. Seedling height is measured at right angles to the slope of the plot. Record natural (as opposed to extended) plant height. Stems that fork at or above ground level are recorded as a single stem. Those that fork below ground are counted as separate stems.
- Do not record saplings or trees that are present on understorey subplots.
- For lianes, herbs and ferns record presence (✓), but not number of individuals, in the height classes used for woody seedlings.
- For non-vascular species (Appendix 10B) record cover class, substrate (rock, soil, living stem, litter, CWD and decay class) and plant type (lichen, hepatic, bryophyte) for the dominant species on each seedling plot. Record species growing on terrestrial substrates (including CWD), but not those originating from canopy litterfall.
- Where non-vascular species are only being sampled on a subset of the understorey subplots (e.g., during the establishment phase of the CMS plot network³) and existing plots use a non-standard numbering system for seedling subplots, use position (as opposed to number) to determine the subplots on which non-vascular species are recorded.
- Note that during the establishment phase of the CMS plot network, the aim is to characterise the major elements of the non-vascular vegetation, rather than establish a complete inventory of the species on each plot.

4.3.2 Saplings

Sapling data are collected from all forest and some shrubland (see Section 4.2.4) plots. The term sapling applies to woody plants (including epicormic shoots and epiphytes) greater than 1.35 m tall but less than 2.5 cm dbh, tree ferns and lianes (see Section 4.1.1 for definitions of epicormic shoots, epiphytes, and lianes). Saplings are counted by species, but they are not tagged.

- For woody tree species the natural (as opposed to extended) height is taken as the measure of height.
- Stems that fork at or above ground level are recorded as a single stem. Those that fork below ground are counted as separate stems.

³ Appendix 4 shows the subset of understorey plots on which non-vascular plants are being sampled during the establishment phase of the CMS programme.

- Lianas are only included in the sapling inventory where they are rooted within the plot. They are classified as saplings where they have a stem diameter >5 mm and <2.5 cm at breast height. The 'breast height' of a liane is taken as 1.35 m from the base of the stem (See also Section 4.1.1).
- Tree ferns are classed as saplings when they have a natural height greater than 1.35 m and a stem height of less than 1.35 m.
- For each subplot (A–P) count the number of saplings of each species, and record these data in the Sapling column of the Stem Diameter/Height record sheet (Appendix 8). Use the $<>$ symbols to denote a sapling record. Record 'none' for subplots where no saplings are present.

4.4 Coarse woody debris

Coarse woody debris (CWD) includes dead standing trees, and fallen logs/branches that have a diameter ≥ 10 cm. Full enumeration is used because some plots contain the majority of their CWD in a few, large, misshapen and partly decayed logs (Stewart & Burrows 1994). These data are collected from all CMS plots, irrespective of whether or not they contain other woody vegetation.

- Locate all dead standing trees and fallen logs greater than 10 cm in diameter within each subplot (A–P) of the 0.04-ha plot, and where appropriate all dead standing trees and fallen logs greater than 60 cm in diameter within the 0.13-ha plot. On plots with a large number of logs/fallen branches, use chalk or some other non-permanent marker to minimise the possibility that CWD items may be missed or double counted.
- Allocate each item of CWD to one of the following decay classes (Allen et al. 1997):
 - 0 Live fallen tree (see Section 4.1.1).
 - 1 Wood hard but not stained, visible growth rings, bark intact.
 - 2 Wood hard but stained, growth rings indistinguishable, bark usually sloughing off. Core of the log remains solid.
 - 3 Advanced wood decay with some loss of original form, bark may remain intact in some species. Wood friable, no solid core.
 - 4 Characters of decay class 3, but loss of shape precludes volume measurement. Do not measure logs of decay class 4.
- The decay class for a log is the one that pertains to the majority of that log. Use a metal probe to determine the presence/absence of a solid core.
- Tag all dead standing trees, tree stumps, and fallen logs in decay classes 1 and 2, and

wherever possible identify each to genus and species level. If both the genus and the species is unknown record 'unknown'. Standing dead trees and logs in decay class 3 should be measured but not tagged.

- For standing dead trees >1.35 m high and ≥ 10 cm dbh, record dbh, height and decay class on the Stem Diameter/Height record sheet (Appendix 8).
- For tree stumps (standing dead stems ≥ 10 cm in diameter and <1.35 m high) record the height and two orthogonal widths (one of which is the largest) at the upper end of the stump on the Coarse Woody Debris record sheet (Appendix 11), and note as a stump.
- For fallen logs (excluding root buttresses), measure the length and two orthogonal widths (one of which is the largest) at both ends. Width measurements should take account of the below-ground portion of the log.
- Only measure those parts of logs that fall within the plot (i.e., logs that are ≥ 10 cm in diameter within the 0.04-ha plot, and >60 cm in diameter within the 0.13-ha plot).
- Where a tree or log dramatically changes shape or decay class along its length, or becomes hollow, measure appropriate segment diameters and lengths to ensure the subsequent volume calculations are accurate. Tag each measured segment of logs in decay classes 1 and 2. Bracket segments belonging to the same item CWD.
- Record data for tree stumps and fallen logs/branches on the Coarse Woody Debris record sheet (Appendix 11).

5 Remeasuring Plots

Experience across a wide a range of New Zealand forest types suggests that a time frame of 5–10 years is generally suitable for detecting changes in stand structure and plant biodiversity (Allen 1993). Where permanent plots are not remeasured for longer periods, location markers and tree tags become overgrown or are lost, and it becomes progressively more difficult to accurately repeat the initial set of measurements.

5.1 Where an existing permanent plot is to be remeasured

- Obtain a copy of the plotsheets (including the problems and corrections sheets) from previous measurements (see Appendices 15A and 15B), map and aerial photo from the NVS databank, to assist in re-locating and remeasuring the plot.
- When remeasuring existing tagged trees use a computer printout of the stem diameter/height data from the previous plot measurement to ensure that:
 - tree/tree fern species are correctly and consistently identified
 - all stems on the plot have been measured/remeasure
 - there are no anomalous measurements (e.g., an incorrectly placed decimal point).
- Once the plot has been found, clear or replace location and plot markers that are obscured, damaged or missing, and update the plot location description. Ensure replacement plot markers are labelled correctly. Where corner and/or seedling plot pegs have been lost, use the plot corner tree markers and tree tag numbers to identify the plot and subplot boundaries.
- Use new datasheets each time the plot is remeasured. **Do not** write data from the current remeasure on photocopies of earlier measurements (see Newell & Baldwin 2000; <http://nvs.landcareresearch.co.nz/>).
- Remeasure all plot description parameters (Section 3.2), and note any points of difference from the previous description.
- Repeat the vegetation description (Section 3.3), again taking care to reconcile differences with records from the previous measurement.
- Where the measured diameter of a tree varies considerably from the previous measurement (e.g., 13.4 cm in 2003 but recorded as 16.3 cm in 1987), note that the current measurement has been checked and is correct. This provides a check that the newly collected diameter record is not a measurement or recording error.
- Corrections to the identification of a species should also be accompanied by a short note confirming that the change was a conscious decision and not a measurement or recording error.
- Where tree tags are overgrown, remove the original tag and use it to retag the tree

(Section 4.1.1) at an adjacent position on the same circumference. Where the tag is damaged beyond repair or partially 'swallowed' by the tree, use a Dymo tapewriter (or similar instrument) to manufacture a replacement metal tag with the same tag number. This ensures the fate of individuals can be tracked over time. **Do not assign a new tag number to a previously tagged stem.** Identify trees that have been retagged in the Notes column of the Stem Diameter/Height record sheet (Appendix 8).

- Where a tree tag cannot be located it is often possible to determine the original tag number using species and dbh data from the previous measurement. Retag the tree using the original number and make a note to this effect in the Comments column.
- Where stems have been consistently tagged above or below breast height, add a CMS tag at breast height, and record the stem diameter at both tag positions. This ensures the correct data is available for carbon calculations while not removing the ability to track demographic changes.
- Where a tree has died since it was last measured, remeasure the stem diameter and note its death in the Comments column of the Stem Diameter/Height record sheet (Appendix 8). For stems ≥ 10 cm dbh also note the decay class (Section 4.4). For stems < 10 cm dbh fold the tag over to indicate a dead stem.
- When tagging new stems, check to see that the new tag numbers do not overlap with the original sequence.
- **Never** reuse tags from dead trees to identify another stem, as this can cause problems when data are being analysed.
- Wherever possible, height measurements (Section 4.1.2) should be taken from the same trees sampled on previous occasions. Recheck height measurements that are inconsistent with values from the previous measurement.
- Use a metal detector to search for 'missing' seedling pegs. In plots that have not been remeasured recently these may be buried as much as 10–15 cm below the current surface. On steep slopes and/or stony substrates pegs frequently 'migrate downslope' or fall out.
- If a seedling peg has fallen out or cannot be found, replace it (use the plot layout sheet from the previous measurement to determine the correct location) and note that the peg has been 're-established', on the current plot layout sheet.
- Repeat shrub biomass, sapling, seedling and other understorey vegetation, and coarse woody debris measurements as described in Sections 4.2–4.4.
- Note that where the vegetation structure or composition has changed substantially since the previous measurement (e.g., regenerating forest), this may necessitate a change in the method of vegetation measurement.

- Errors or inconsistencies that are detected in existing plot data in the NVS databank, are recorded on a 'problems sheet' (Appendix 15A) for checking when the plot is next remeasured (Newell & Baldwin 2000; <http://nvs.landcareresearch.co.nz/>). Where field parties are able to solve the identified problem, the 'problem sheet' should be returned to The Curator, National Vegetation Survey databank, Landcare Research, PO Box 69, Lincoln 8152, who will update the databank records. The resulting changes to electronic datafiles are recorded on a 'corrections sheet' (Appendix 15B), which is archived with the plot sheets. The 'corrections sheet' is a record of all changes that have been made to the original data.

5.2 Where an existing NVS plot is to be remeasured

NVS plot protocols have changed slightly over the years and at times have been subject to differences in interpretation. When remeasuring an existing NVS plot it is important to follow the original measurement protocols to maintain the integrity of the time-series element of the dataset.

- Only replace existing plot markers or seedling pegs where they are damaged or cannot be found. Note any replacements on the Plot Layout sheet (Appendix 4).
- Do not adjust plot axes that are found to be $>$ or < 20 m, as this will invalidate comparisons with earlier measurements.
- Some early NVS plots labelled the plot corners 'A', 'B', 'C', 'D' rather than 'A', 'D', 'M', 'P'. Where this lettering system is encountered retain the status quo and note this on the Plot Layout sheet.
- Where a previously tagged tree is found to be outside the plot boundary or a non-tagged tree is found within the plot boundary, measure the tree and record its position in the Notes column. Where this involves a marginal call, leave the situation as you found it.
- Several different systems for numbering seedling pegs have been used since 20×20 -m plots were first introduced (Fig. 5). As for the plot corners, please retain the original numbering system, and note any differences from the CMS seedling subplot layout on the Plot Layout sheet (Appendix 4). Be sure you have correctly identified the existing subplot numbering sequence before replacing missing plot pegs or beginning to remeasure seedling subplots. It is frequently possible to identify the numbering sequence from as few as four labelled seedling pegs that are widely spaced across the 0.04-ha plot.

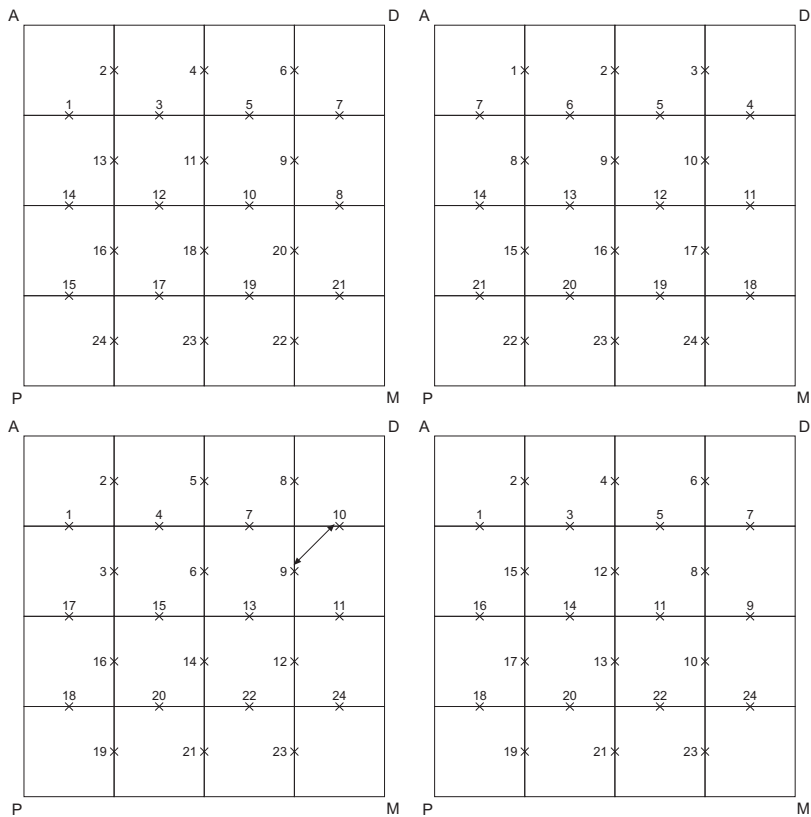


Figure 5. Alternative seedling plot layouts found on some older NVS plots.

6 Conclusions

Over the course of a long-term monitoring programme there is often pressure to change sampling strategies and/or data collection protocols, to accommodate changed objectives and/or new and improved methodologies. Where changes are proposed it is important to ensure that they do not compromise the outcomes of both the original and the revised monitoring programmes. Where a new or modified measurement protocol is adopted, both methods should be used the next time the plots are remeasured. This allows the effects of the change in protocol to be assessed and incorporated into existing long-term datasets.

7 Quality Control Procedures

Quality Control (QC) is the system of routine technical activities that ensure the quality of the inventory that is being developed, in this case the provision of reliable datasets for the estimation of carbon stocks and plant biodiversity.

For CMS forest and shrubland plots the QC procedures are as follows:

Prior to the commencement of fieldwork:

- Check that all equipment required is present (Appendix 3) and in good working order (e.g., altimeter calibrated, spring balance reading correctly).
- Ensure the party has the best available information to locate/re-locate the CMS plot.
- Ensure the party is fully informed of all plot measurement methodologies.

Before the party leaves the plot check that:

- Plant specimens – labelled and securely stored in a field plant press.
- Data sheets – all fields completed and no obvious anomalies.
- Shrub biomass samples are correctly packaged and labelled.

En route to home base ensure:

- Secure storage of samples and data sheets at the field base.
- Adequate drying of plant specimens.
- Sample storage as per the requirements specified in this manual.
- Where field crews are in the field for periods of more than 2 weeks, samples and data sheets should be returned to home base by tracked mail or equivalent courier service.

At home base check that:

- Plant specimens are referred to relevant taxonomist for identification.
- Sample processing equipment is available and in good working order (e.g., drying ovens and weighing balances correctly calibrated).
- Data sheets – plant identifications corrected, and data processing information added.
- Electronic data entry and checking routines completed, data sheets and plant samples archived in an appropriate herbarium (e.g., CHR) or databank (e.g., NVS).
- Corrected problems sheets for existing NVS plots sent to the Curator, NVS databank, Landcare Research, P.O. Box 69, Lincoln 8152.

8 Data Storage

Routine archiving of data is an essential part of any monitoring system. Because data from previous sampling periods form part of a time-series, it is not possible to recover data that have been lost. For these reasons it is important to ensure that datasets are archived in secure, preferably climate-controlled, insect- and fire-proof storage.

For the NZCMS, originals of all plot sheets, maps, aerial photographs, location diagrams and computer datafiles are archived in the National Vegetation Survey databank managed by Landcare Research, Lincoln. This archive incorporates data from vegetation surveys carried out by the former New Zealand Forest Service (now Landcare Research, Forest Research and the Department of Conservation), the National Forest Survey (1946–55), and the Protected Natural Areas Programme. The databank, which comprises hard copy, computerised datafiles, and analysis software for a range of standardised monitoring methodologies, can be accessed via the Internet at <http://nvs.landcareresearch.co.nz/>.

9 Acknowledgements

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10 References

- Allen, R.B. 1992: RECCE: an inventory method for describing New Zealand vegetation. Ministry of Forestry, Christchurch. *Forest Research Institute Bulletin* 181. 25 p.
- Allen, R.B. 1993: A permanent plot method for monitoring changes in indigenous forests. Manaaki Whenua – Landcare Research, Christchurch. 35 p.
- Allen, R.B.; Clinton, P.W.; Davis, M.R. 1997: Cation storage and availability along a *Nothofagus* forest development sequence in New Zealand. *Canadian Journal of Forestry Research* 27: 323–330.
- Burrows, L.E. 2000: Guidelines for fixing co-ordinates of NVS (National Vegetation Survey) databank plots using GPS (Global Positioning System). Pp. 53–59 in Allen, R.; Wiser, S.; Burrows, L.; Brignall-Theyer, M. 2000: Silvicultural research in selected forest types: a black beech forest in Canterbury. Unpublished report (LC0001/01) for the Ministry of Agriculture and Forestry, Wellington.
- Coomes, D.; Beets, P. 1999: Monitoring carbon in indigenous forests and shrublands: a test of methods and logistics along a South Island transect. Unpublished report (JNT9899/124) for the Ministry for the Environment, Wellington. 56 p.
- Coomes, D.A.; Allen, R.B.; Scott, N.A.; Goulding, C.; Beets, P. 2002. Designing systems to monitor carbon stocks in forests and shrublands. *Forest Ecology and Management* 164: 89–108.
- Hall, G.M.J. 1992: PC-RECCE: vegetation inventory data analysis. Ministry of Forestry, Christchurch. *Forest Research Institute Bulletin* 182. 108 p.
- Lynn, I.H.; Crippen, T.F. 1991: Rock type classification for the New Zealand Land Resource Inventory. *DSIR Land Resources Scientific Report* 10. 123 p.
- McNab, W.H. 1993: A topographic index to quantify the effect of mesoscale landform on site productivity. *Canadian Journal of Forest Research* 23: 1100–1107.
- Mueller-Dombois, D.; Ellenberg, H. 1974: Aims and methods of vegetation ecology. John Wiley and Sons, New York. 547 p.
- Newell, C.L.; Baldwin, A.B. 2000: Data-standard guidelines for improving the quality of permanent plot data archived in the National Vegetation Survey databank – first approximation. Unpublished report (LC9900/140) for the Foundation for Research, Science and Technology, Wellington. 52 p.
- Payton, I.J.; Pekelharing, C.J.; Frampton, C.M. 1999: Foliar Browse Index: a method for monitoring possum (*Trichosurus vulpecula*) damage to plant species and forest communities. Landcare Research, Lincoln. 62 p.
- Stewart, G.H.; Burrows, L.E. 1994: Coarse woody debris in old-growth temperate beech (*Nothofagus*) forests of New Zealand. *Canadian Journal of Forest Research* 24: 1989–1996.

- Wiser, S.K.; Bellingham, P.J.; Burrows, L.E. 2001: Managing biodiversity information: development of New Zealand's National Vegetation Survey databank. *New Zealand Journal of Ecology* 25: 1–17.

11. Appendices

- Appendix 1. Metadata sheet
- Appendix 2. Recce plot sheet
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 - (B) species composition
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 - (B) corrections sheet

Appendix 1 Metadata sheet

NEW ZEALAND CARBON MONITORING SYSTEM
METADATA SHEET

CMS PLOT IDENTIFIER: AM135
 NVS PLOT IDENTIFIER (if applicable): n.a.
 CMS SOILS PLOT: YES/NO
 LANDCOVER/LAND-USE CLASS: shrubland (shrubland)
 CATCHMENT/AREA/REGION: Tasman Valley
Mt Cook National Park, Canterbury
 TOPOMAP: H36 Mount Cook
 AERIAL PHOTO: -
 GPS REFERENCE: (Easting) 2280054
 (Northing) 5717076
 GPS (make and model): Garmin 12 XL
☒ 3D fix (delete one)
☒ Single/Averaged position fix (delete one) ± 6.4 m
 LAND TENURE OF PLOT: Conservation Estate (National Park)
 PERMISSION CONTACTS:
 For plot:
 Name: Ray Bullinger
 Address: Dept of Conservation, P.O. Box 5, Mt Cook
 Phone number: 03 435 1819
 Fax number: 03 435 1895
 Email: rbullinger@doc.govt.nz
 Method of contact by CMS team: phone
 Name of CMS contact person: Clare Newell
 Gate keys necessary: No
 Planned access date: 11 November 2002
 Notes: Because the plot site is very close to Tasman Valley Road, please let
DOC (Mt Cook) staff know when the field team proposes to measure the plot.
 TIME ALLOCATION:
 Access
 Town/area where team based for plot: Twizel
 Drive to plot: 45 mins
 Fly to plot: n.a.
 Walk to plot: 2-3 mins
 Search time to relocate NVS plot: n.a.
 Soils plots
 Fine Woody Debris/LFH sampling: 1 hour
 Mineral soil sampling: 2 1/2 hours
 Soil profile description: 1 1/2 hours
 Total time to measure soils plot: 5 hours (2 people)
 Vegetation plots
 Plot layout: 1 hour (6 people)
 Recce description: 1 1/2 hours
 Stem diameters: n.a.
 Tree heights: n.a.
 Coarse woody debris: n.a.
 Understorey plots: 7 1/2 hours
 Individual/Continuous shrubs: 5 1/2 hours
 Total time to measure veg plot: 8 1/2 hours (4 people)
 ADDITIONAL COMMENTS ABOUT PLOT:
Plot near the edge of steep roadside bank. Red perimolot marker close to
corner P is visible from roadside.

¹Complete where information is available

Appendix 2A Recce plot sheet – site description

NEW ZEALAND CARBON MONITORING SYSTEM
RECONNAISSANCE (RECCE) PLOT SHEET

CMS PLOT IDENTIFIER: J158
 CATCHMENT/AREA/REGION: Rum River
Brookshaw Sound, Fiordland
 TOPO MAP: B12, Secretary Island
 GPS REFERENCE (Easting): 2049039
 (Northing): 5533053
 GPS (make and model): Garmin 12XL
~~QZSD fix (delete one)~~ ~~Shape~~ (Averaged position (delete one)) = 9.5m

LANDCOVER/LAND-USE CLASS: forest (forest)
 AERIAL PHOTO: —
 MEASURED BY: Larry Burrows
Alvin Miller
 RECORDED BY: Alvin Miller
 DATE: 9 April 2002

SITE DESCRIPTION

SITE DESCRIPTION		SURFACE STABILITY		HORIZON ANGLES (°)		SOIL DEPTH (cm)	
ALTITUDE (m)	260	ROCK ON SURFACE		(YES) NO		(record + or -)	
PHYSIOGRAPHY	Ridge, Face, Gully, Terrace	BEDROCK ON SURFACE		(YES) NO		N	A
ASPECT (S-360°)	30	BEDROCK %		40		NE	D
SLOPE (°)	32	BROKEN ROCK %		10		E	M
PARENT MATERIAL	on-site granite	SOIL %		50		SE	P
	mapped gneiss	SIZE OF LOOSE ROCK		30 or >30 cm		S	C
DRAINAGE	Good, Moderate, Poor	ALLUVIAL (COLLUVIAL) VULCANISM		SW		+60	
CULTURAL	None, Fire, Logged, Cleared, Mined, Grazed, Tracked			W		+38	
				NW		+15	
						GROUND COVER %	
						Veg	
						Moss	
						Litter	
						Bare Ground	
						Rock	

APPROACH

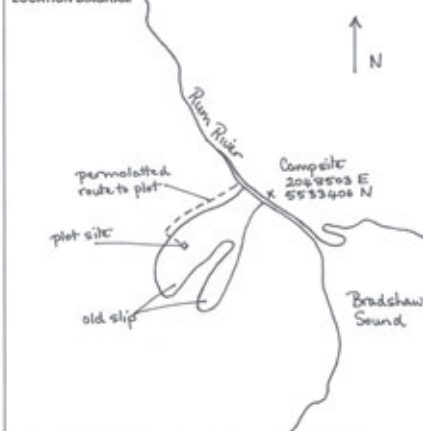
Plot site is on an old slip on the true right of Rum River, approx 1 km from the river mouth. Percolated route to the plot starts at the mouth of the small creek immediately upvalley of the slip. Route travels up the side of the slip through silver beech forest to an altitude of 260 m. Then follows the contour to the plot site, which is almost midway across the slip.

NOTES (including cultural)

Canopy dominated by CO2Rho, CYA smi, & SCH dig. Dense understory of POLWES. Creek runs through the centre of the plot. Deer browse and antler rub on FULCCE and SCH dig.

*Complete where information is available

LOCATION DIAGRAM



Appendix 2B Recce plot sheet – species composition

CMS PLOT IDENTIFIER: S160 MEASURED BY: Meredith McKay, Matt Renner
 DATE: 24 April 2002 RECORDED BY: Meredith McKay

SPECIES COMPOSITION

Cover Class: <1% = 1; 1-5% = 2; 6-25% = 3; 26-50% = 4; 51-75% = 5; 76-100% = 6
 1% = 2 X 2 m; 5% = 4 X 5 m.

	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5	Tier 6	
	25 m +	25 m - 12 m	12 m - 5 m	5 m - 2 m	2 m - 0.3 m	<0.3 m	<0.3 m
						vascular	non-vascular terrestrial
Height		21-24 m	6-12 m				
Cover		5	4	3	2	3	
Diameter		60-100 cm					PTY aci
							WEY coc
		NOT men 2	✓ 1	✓ 1	✓ 1	✓ 1	DIC rob
		NOT cli 4	✓ 2	✓ 2	✓ 1	✓ 3	HYP cup
			NOT fus 2	✓ 1	✓ 1	✓ 1	PLA nut
			CAR ser 2	✓ 1	✓ 1	✓ 1	PLA cir
				PSE cra 1	—	—	DIS cri
					MYR dri 2	✓ 1	LEP gam
					COP rha 1	✓ 1	THU fur
					COP can 1	✓ 1	
					COP tay 1	✓ 1	
					COP hairy 1 [Ⓢ]	—	
					RUB cis 1	—	
					NIC ave 1	✓ 1	
					POL vis 1	✓ 1	
						UNC fer 1	
						UNC fin 1 [Ⓢ]	
						NER vil 1	terrestrial on
						BLE pen 1	woody debris
						CAR deb 1	
						HYM mal 1	RIC lob
						HYM small 1 [Ⓢ]	FRU ate
						COR tri 1	FRU fol
							PHY bic
							PTY aci
							SCH cil
							WEY coc
							HET bil
							LEP sp. [Ⓢ]

BROWSE

Plant species	COP tay						
Herbivore	deer						
Severity	L (M) H	L M H	L M H	L M H	L M H	L M H	L M H

FAUNA (mammal, bird, reptile, invertebrate)

tomtit, kakariki, robin
deer pellets in vicinity of plot.

Appendix 3 Equipment required to establish and measure forest and shrubland plots

List of equipment required to establish and measure forest and shrubland plots

Plot measurement equipment

Forest & Shrubland data collection manual
 Plant identification texts
 Foliar Cover Scale
 20 × 20-cm frame
 Topographical maps and aerial photos
 Geological survey map
 Global Positioning System unit
 Altimeter
 Abney level or hypsometer
 Vertex or Suunto hypsometer
 Compass
 Soil probe
 Hip chain
 Spring balance (5 & 20 kg)
 Lightweight metal detector to re-locate plot pegs
 20-m tapes
 Steel tapes (3 & 8 m)
 Pruning saw, secateurs, shears, loppers
 Large plastic bags for weighing shrub biomass
 Extendable height pole – tree/shrub heights
 Plant press
 Dymo tapewriter (or similar instrument)
 Stem diameter tape
 20-m nylon cords
 Seedling plot string (49 cm long)
 Hammer (for nailing tree tags)
 Tree-tag dispenser
 Binoculars

Consumable items

Clipboards, pens, pencils
 Waterproof labels
 Large-sized aluminium pegs (45 × 0.7 cm)
 Small-sized aluminium pegs (30 × 0.5 cm)
 Nails (40-mm galv. flathead) – to tag trees
 Nails (75-mm galv. flathead) – to tag tree ferns
 Aluminium tree tags (numbered series)
 Metal tape for Dymo tapewriter
 ‘Permolat’ flagging
 Plastic bags for sample storage
 Self-sealing plastic bags – moisture content samples
 Paper bags for non-vascular plant specimens
 Hipchain cotton
 Batteries for GPS and vertex hypsometer
 Chalk

Data entry sheets

Metadata sheet: all plots
 Reconnaissance (Recce) plot sheet (site description and species composition): all plots
 Plot Layout sheet: all plots
 Stem Diameter/Height record sheet: forest plots and some shrubland plots
 Discrete Shrub record sheet: all plots where discrete shrub cover is measured
 Continuous Shrub Cover record sheet: all plots where continuous shrub cover is measured
 Shrubland Plot Layout sheet: all shrubland plots
 Shrub Biomass Harvest record sheet: all shrubland plots
 Understorey Record sheet: all plots
 Coarse Woody Debris record sheet: all plots with items of coarse woody debris

Appendix 4 Plot Layout sheet

NEW ZEALAND CARBON MONITORING SYSTEM

PLOT LAYOUT SHEET

CMS PLOT IDENTIFIER: CU86 MEASURED BY: Pete Shaw, Matt Calder

DATE: 30 August 2002 RECORDED BY: Karen Riddell

Corner marker on N4R sal. 0.7m from A at 230°

EXT

Key
○ non-vascular subplots
Corner marker on L18 bid. 2.7m from D at 162°

Corner marker on WEI sal. 3.6m from M at 110°

Corner marker on N4R sal. 3.8m from P at 305°

RECORD ON THE ABOVE DIAGRAM

1. Lengths A-D, D-M, M-P, A-P
2. Position (distance and direction) of percolat strips used to locate plot corners
3. Location of trees & CWD >60 cm diameter in the large circular plot
4. Location of seedling subplots with distances
5. Location of soil sample plots (where appropriate)

Appendix 5 Landcover/Land-use class definitions

Landcover/Land use class definitions used by version one of the LandCover DataBase (LCDB1).

Artificial landscapes

Urban – built up areas. Includes any contiguous group of buildings larger than the minimum mapping area of one hectare.

Mines, gravel pits and dump sites

Urban open space – e.g., sports fields, parklands etc.

Cultural landscapes

Primarily pastoral – exotic pasture, enclosure distinguishes this from grasslands. Includes arable land.

Primarily horticultural – orchards, kiwifruit and market gardens.

Planted forest – plantation or exotic forest inclusive of recent replanting, e.g., *Pinus radiata*, *Pseudotsuga menziesii*, *Eucalyptus* spp.

Shelterbelts – major shelterbelts (visible on [satellite] imagery).

Riparian planting – willows and poplars.

Natural landscapes

Grassland – tussock, or unenclosed grassland.

Shrubland – Woody vegetation in which the cover of shrubs and trees in the canopy is >20% and in which shrub cover exceeds that of any other growth form or bare ground. Shrubs are woody plants, e.g., manuka, kanuka, matagouri, gorse, broom, hawthorn.

Indigenous forest – forest cover dominated by indigenous tall forest canopy species.

Bare ground – non-pastoral exposed soil and rock.

Inland wetlands – inundated by fresh water.

Coastal wetlands – inundated by salt water.

Coastal sands – beach sands and dunes.

Inland water – lakes, ponds and rivers.

Mangrove – sea level mangrove swamp land.

Appendix 6 Foliage Cover Scale



Appendix 7 Non-standard codes for vascular plants in the New Zealand flora

Non-standard six-letter codes for vascular plants in the New Zealand flora

Rules for constructing six-letter plant species codes

1. The plant species code consists of the first three letters of the genus (upper case), followed by the first three letters of the species or subspecies (lower case). For example: Black beech (*Nothofagus solandri* var. *solandri*) is recorded as NOTsol, and mountain beech (*Nothofagus solandri* var. *cliffortioides*) is recorded as NOTcli.
2. Where only the genus is known (e.g., *Metrosideros*) use the first six letters (upper case) (e.g., METROS).
3. For hybrids with a recognised hybrid name (e.g., *Coprosma cunninghamii* = *Coprosma propinqua* x *robusta*), the code consists of the first three letters of the genus (upper case) followed by an x (to denote the hybrid status of the plant) and the first two letters of the hybrid name. Thus the six-letter code for *Coprosma cunninghamii* is COPxcu.
4. For hybrids without a recognised hybrid name (e.g., *Nothofagus solandri* var. *cliffortioides* x *N. fusca*), the code consists of the first three letters of the genus (upper case) followed by the first letter of the each putative parent (lower case) separated by an x. Thus the six-letter code for the mountain x red beech hybrid is NOTcxf.
5. For species listed in the New Zealand Plant Names database but not formally described (e.g., *Coprosma* species (*a*)), the code consists of the first three letters of the genus (upper case) followed by the letter used to identify the informal species (lower case) (e.g., COPlt).
6. Where application of these rules doesn't provide an unambiguous species code, either use the following list of non-standard six-letter codes or write the plant name in full. DO NOT create adhoc non-standard plant species codes, as these are likely to be misinterpreted by persons who are not familiar with the dataset in question.

Non-standard six-letter plant species codes

<i>Aciphylla similis</i>	ACIsim
<i>Aciphylla simplex</i>	ACIsmp
<i>Agrostis perennans</i>	AGRper
<i>Agrostis personata</i>	AGRpes
<i>Asplenium colensoi</i>	ASPcol
<i>Asplenium hookerianum</i> var. <i>colensoi</i>	ASPPhvc
<i>Asplenium flaccidum</i>	ASPfla
<i>Asplenium flabellifolium</i>	ASPflb
<i>Beilschmiedia tawa</i>	BEItaw
<i>Beilschmiedia tawaroa</i>	BEIroa
<i>Brachyscome</i> species	BRACHY
<i>Brachyglottis</i> species	BRACHG
<i>Carex coriacea</i>	CARcor
<i>Carmichaelia corrugata</i>	CARcog
<i>Cardamine corymbosa</i>	CARcoy
<i>Carex flaviformis</i>	CARfla
<i>Carex flagellifera</i>	CARfgl
<i>Carmichaelia flagelliformis</i>	CARflg
<i>Carex kirkii</i>	CARkir
<i>Carmichaelia kirkii</i>	CRMkir
<i>Carmichaelia petrei</i>	CARpet
<i>Carex petrei</i>	CARptr
<i>Carex virgata</i>	CARvir
<i>Carmichaelia virgata</i>	CARvig
<i>Celmisia gracilentia</i>	CELgra
<i>Celmisia graminifolia</i>	CELgrm
<i>Celmisia petrei</i>	CELpet
<i>Celmisia petiolata</i>	CELpel

<i>Cerastium vulgatum</i>	CERvul	
<i>Cerastium fontanum</i> subsp. <i>vulgare</i>	CERfsv	
<i>Coprosma tenuicaulis</i>	COPtec	
<i>Coprosma tenuifolia</i>	COPtef	
<i>Cyathea colensoi</i>	CYAcot	
<i>Cyathodes colensoi</i>	CYTcol	now LEUcol
<i>Cyathea dealbata</i>	CYAdea	
<i>Cyathodes dealbata</i>	CYTdea	
<i>Cyathodes empetrifolia</i>	CYTemp	
<i>Cyathodes fasciculata</i>	CYTfas	now LEUfas
<i>Cyathodes fraseri</i>	CYTfra	now LEUfra
<i>Cyathodes juniperina</i>	CYTjun	
<i>Cyathodes parviflora</i>	CYTpar	now LEUpar
<i>Cyathodes pumila</i>	CYTpum	
<i>Cyperus eragrostis</i>	CYPera	
<i>Cyperaceae</i> species	CYPspp	
<i>Deschampsia chapmanii</i>	DEScha	
<i>Deschampsia</i> species	DESchm	
<i>Dracophyllum prunum</i>	DR Apro	
<i>Dracophyllum prostratum</i>	DR Aprs	
<i>Drosera peltata</i> subsp. <i>auriculata</i>	DROpel	
<i>Gaultheria novaezealandiae</i>	GAUnov	
<i>Gaultheria depressa</i> var. <i>novaezealandiae</i>	GAUdvn	
<i>Hebe stricta</i>	HEBstr	
<i>Hebe strictissima</i>	HEBsts	
<i>Hebe tetragona</i>	HEBtet	
<i>Hebe tetrasticha</i>	HEBtrr	
<i>Hydrocotyle montana</i>	HYDmon	
<i>Hydrocotyle novae-zeelandiae</i> var. <i>montana</i>	HYDnzm	
<i>Myosotis pygmaea</i> var. <i>druceii</i>	MYOpyd	
<i>Ourisia macrocarpa</i> var. <i>macrocarpa</i>	OURmac	
<i>Ourisia macrophylla</i> var. <i>macrophylla</i>	OURmap	
<i>Pittosporum crassicaule</i>	PITcra	
<i>Pittosporum crassifolium</i>	PITcrf	
<i>Plantago lanceolata</i>	PLAlan	
<i>Plantago lanigera</i>	PLAlng	
<i>Pseudowintera colorata</i>	PSEcol	
<i>Pseudopanax colensoi</i>	NEOcol	
<i>Pseudopanax</i> species	PSEUDO	
<i>Pseudognaphalium</i> species	PSEUDG	
<i>Pseudowintera</i> species	PSEUDW	
<i>Pteridium aquilinum</i> var. <i>esculentum</i>	PTEqu	
<i>Raoulia subulata</i>	RAOsub	
<i>Raoulia subsericea</i>	RAOsb	
<i>Schizaelema nitens</i>	SCHnit	
<i>Schoenus nitens</i> var. <i>concinus</i>	SCHnvc	
<i>Schoenus nitens</i> var. <i>nitens</i>	SCHnvn	
<i>Stellaria gracilenta</i>	STEgra	
<i>Stellaria graminea</i>	STEgrm	
<i>Stenostachys gracilis</i>	STEgrc	
<i>Trichomanes strictum</i>	TRIstr	
<i>Trifolium striatum</i>	TRIstt	
<i>Triglochin striatum</i>	TRISTA	

Appendix 8 Stem Diameter/Height record sheet

NEW ZEALAND CARBON MONITORING SYSTEM
STEM DIAMETER/HEIGHT RECORD SHEET

CMS PLOT IDENTIFIER: BN 98 MEASURED BY: Rob Fraser
 DATE: 22 October 2002 RECORDED BY: Pete Shaw

Record measurements for standing CWD on this sheet: diameter, height, angle of lean, and decay class

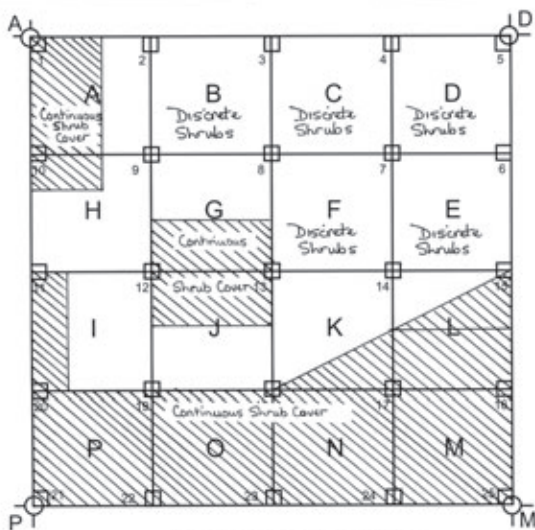
Sub-plot	Species	Tag No.	Diameter	Height	Angle of lean	Decay class	No. of saplings	Notes
A	Pod hol	W21126	35.6	18.2				
A	COP lin	W21127	5.3	4.6				
<A>	Rub cis						II = 3	
<A>	NOT men						I = 1	
<A>	COP lin						III = 5	
<A>	COP lin						I = 1	epicormic on W21127
B	NOT men	W0895	32.8	12.5		2		dead standing stem
B	Pod hol	W21128	19.4	10.3	-30°			
B	Pod hol	W21129	3.9					
B	GRI lit	W21130	4.6					epiphyte on W21128
B	COP lin	W21131	10.3					
B	COP lin	W21132	15.5					
	COP lin						II = 2	
	COP for						III = 6	
C	GRI lit	W21133	10.1	8.1	+40°			
C	NET umb	W0896	26.4	6.5		2		dead standing stem
C	NET umb	—	15.0	4.1		3		dead standing stem
C	GRI lit	W21134	6.0					
C	GRI lit	W21135	9.3					
C	GRI lit	W21136	20.6	12.6				
<C>	none							
EXT	NOT fus	W21137	71.7	23.2				adj. to Quad M.
EXT	NOT fus	W21138	60.3	18.3				adj. to Quad D.
EXT	NOT fus	W0897	61.5	3.7				dead standing stem

¹ When measuring tree height estimate the angle of lean to the nearest 10° for trees with a lean of >20°. Record as + where the lean is towards the point from which the tree height is measured, and - if the tree is leaning away from this position. Do not record an angle of lean where tree height is measured at right angles to the angle of lean.

Appendix 9 Shrubland Plot Layout sheet

NEW ZEALAND CARBON MONITORING SYSTEM
SHRUBLAND PLOT LAYOUT

CMS PLOT IDENTIFIER: BR 116 MEASURED BY: Meredith McKay
 DATE: 18 December 2002 RECORDED BY: Tom Orcheston



4x1 m continuous biomass
harvest plot is 5.0 m from
corner M, on a bearing of 125°

On the above diagram

1. Illustrate the extent of shrub cover within each subplot, and identify as continuous or discrete
2. Record the location of the destructive shrub sample site(s)

FOR SUBPLOTS WITH CONTINUOUS SHRUB COVER (>30cm height)

Record the dimensions of the continuous shrub cover illustrated on the above diagram. Keep the measurements simple (e.g., rectangles, squares, right-angled triangles). Record dimensions to the nearest 0.1 m.

SUB-PLOT	SHRUB COVER DIMENSIONS	SUB-PLOT	SHRUB COVER DIMENSIONS	SUB-PLOT	SHRUB COVER DIMENSIONS
A	3.2 x 5.0 m rectangle	G	2.4 x 5.0 m rectangle	M	5.0 x 5.0 m square
B	discrete shrubs	H	3.2 x 1.5 m rectangle	N	5.0 x 5.0 m square
C	discrete shrubs	I	1.0 x 5.0 m rectangle	O	5.0 x 5.0 m square
D	discrete shrubs	J	2.3 x 5.0 m rectangle	P	5.0 x 5.0 m square
E	discrete shrubs	K	2.5 x 5.0 m rt angle triangle		
F	discrete shrubs	L	2.4 x 5.0 m rt angle triangle + 2.6 x 5.0 m rectangle		

Appendix 10A Understorey record sheet – vascular plant species

NEW ZEALAND CARBON MONITORING SYSTEM
UNDERSTOREY RECORD SHEET
VASCULAR PLANT SPECIES

CMS PLOT IDENTIFIER: BJ 93 MEASURED BY: Karen Riddell
DATE: 30 October 2002 RECORDED BY: Rhodri Harfoot

Sub-plot No.	Species	<15 (°)	16-45	46-75	76-105	106-135
1	NER d.c	✓				
1	LlB pul	✓				
1	BLE pro	✓	✓			
1	RIP sca	✓				
1	GRI lit	✓	2			
1	DIA nig		✓			
1	COP rha	✓				
1	HED arb	✓				
1	STI can		✓			
2	DAC cup	✓				
2	TRE tan	✓				
2	PLU tax		1			
2	UNC rup	✓	✓			
2	BLE pro	✓	✓			
2	NOT men			2		
2	MYR sal	✓				
2	DIA nig	✓				
2	NET umb	✓				
3	MYR sal	✓				
3	NOT men	✓				
3	GRI lit	✓				
3	AST ner		✓			
3	QU ser			2	1	1
3	HYM dem	✓				
4	GRA bil	✓				
4	WEI rac				1	
4	PHY alp		1	2		
4	LEU fas		1			
4	MYR sal	✓				
5	MYR sal	✓				
5	BLE fra	✓	✓			
5	NOT men	✓				
5	COP rha	✓				
5	QU ser	✓	1	2	1	
5	NEL smi		1		1	
5	COP foe	✓				
5	GRI lit	✓				
5	ALS mac		1			
6	NOT men	✓	1			
6	BLE pro	✓	✓			
6	POD hal				1	
6	NEO pad	✓	4	2		
6	UNC rup	✓				
6	MYR sal					1
6	AST ner		✓			
6	LlB pul	✓				
6	BLE fra	✓				
6	COP foe	✓				
6	LIN tri	✓				
7	BLE pro		✓			
7	NOT men	✓		1		
7	GRI lit	✓				
7	COP cal	✓				
7	MYR sal	✓				
7	NEO pad		2	1		

Appendix 10B Understorey record sheet – non vascular plant species

NEW ZEALAND CARBON MONITORING SYSTEM
UNDERSTOREY RECORD SHEET
NON-VASCULAR PLANT SPECIESCMS PLOT IDENTIFIER:
DATE:AI 133
23 March 2003MEASURED BY:
RECORDED BY:Thomas Paul
Thomas Paul

Sub-plot No.	Species	Cover Class ¹	Sub-Strate ²	Plant Type (L/H/B) ³	Sub-plot No.	Species	Cover Class ¹	Sub-Strate ²	Plant Type (L/H/B) ³
1	Zoopsis argentea	2	CWD 4	H	22	Chiloscyphus chlorophyllus	3	CWD 1	H
1	Bazzania adnixa	2	CWD 2	H					
1	Distichophyllum patchellum	2	CWD 1	B	22	Wijkia extenuata	2	CWD 1	B
					22	Hypnodendron comatum	2	Soil	B
4	Heteroscyphus tricanthus	1	Soil	H					
4	Cyathophorum bulbosum	1	Soil	B					
4	Camptochaete arbuscula	1	Soil	B					
7	Lopidium concinnum	1	Soil	B					
7	Hypnodendron comatum	5	Soil	B					
7	Cyathophorum bulbosum	1	Soil	B					
10	Chiloscyphus chlorophyllus	2	CWD 3	H					
10	Sticta lacera	2	CWD 1	L					
10	Plagioschila ramosissima	2	Soil	H					
13	Hypnodendron comatum	2	Soil	B					
13	Heteroscyphus coalitii	2	CWD 2	H					
13	Wijkia extenuata	3	Soil	B					
16	Rhizogonium distichum	1	CWD 3	B					
16	Leucobryum candidum	1	CWD 3	B					
16	Zoopsis argentea	1	CWD 3	H					
19	Sticta lacera	2	CWD 1	L					
19	Wijkia extenuata	2	CWD 1	B					
19	Pyrrhomonium aciculare	3	CWD 4	B					

¹ Cover Class: <1% = 1; 1-5% = 2; 6-25% = 3; 26-50% = 4; 51-75% = 5; 76-100% = 6.On a seedling subplot 1% = 9.8cm² or 3.1x3.1cm; 5% = 22cm² or 4.7x4.7cm² Substrate: Rock, Soil, CWD + decay class, Live tree, or Litter³ Plant type: Lichen, Hepatic, Bryophyte (L, H, B)

Appendix 11 Coarse Woody Debris record sheet

NEW ZEALAND CARBON MONITORING SYSTEM
COARSE WOODY DEBRIS RECORD SHEETCMS PLOT IDENTIFIER: CI 57
DATE: 23 September 2002MEASURED BY: Rhodri Harfoot
RECORDED BY: Chris Bycroft

Sub-plot	Species	Tag No.	Decay Class	Standing/ Fallen S/F	Hollow Y/N	Length (m)	Large End (cm)		Small End (cm)	
							Width 1	Width 2	Width 1	Width 2
A	unknown	—	3	F	N	1.1	14	19	17	16
B	VITluc	W1 7376	2	F	N	2.1	50	37	39	29
B	VITluc	—	3	F	N	0.6	22	17	15	14
B	RHO sap	—	3	F	N	1.7	19	18	10	10
C	none									
D	LITcal	W1 7377	1	F	N	1.7	31	20	17	14
D	LITcal	W1 7378	1	F	N	1.1	13	12	10	10
D	LITcal	W1 7379	1	F	N	1.2	11	10	10	10
D	unknown	—	3	F	N	2.6	15	13	12	10
E	CYA cur	—	3	F	N	3.1	27	17	17	15
E	RHO sap	—	3	F	N	3.3	25	21	17	14
F	VITluc	W1 7380	2	F	N	3.8	55	31	42	40
F	VITluc	W1 7381	2	F	N	0.6	24	22	14	10
F	CYA med	W1 7382	2	F	Y	2.8	27	22	12	12
F	VITluc	—	3	stump	N	1.3	120	114	—	—
G	CYA dea	W1 7383	1	F	N	5.0	21	21	20	19
G	GR lit	W1 7384	1	F	N	2.8	30	23	10	10
G	CYA cur	W1 7385	2	S	Y	4.1	14	12	10	10
H	none									
I	CYATHE	—	3	F	Y	2.4	18	15	17	12
I	unknown	—	3	F	Y	1.2	16	15	13	11
I	VITluc	W1 7386	2	F	N	2.0	115	96	85	62

¹ Measured to one decimal place

Appendix 12 Discrete Shrub record sheet

NEW ZEALAND CARBON MONITORING SYSTEM
DISCRETE SHRUB RECORD SHEETCMS PLOT IDENTIFIER: AM 157
DATE: 8 November 2002MEASURED BY: Hazel McKay
RECORDED BY: Hazel Broadbent

NON-DESTRUCTIVE PLOT MEASUREMENTS

Sub-plot	Species	Width 1 (m)	Width 2 (m)	Height (m)	Sub-plot	Species	Width 1 (m)	Width 2 (m)	Height (m)
A	Distou	2.6	1.9	1.7	E	MELalp	0.6	0.7	0.4
A	Distou	0.8	0.8	1.2	E	MELalp	1.2	0.9	0.5
A	Distou	1.0	1.2	1.3	E	COPrig	0.4	0.3	0.5
A	Distou	2.2	2.0	1.7	E	COPrig	0.2	0.3	0.5
A	ROSrub	0.8	0.6	0.9	E	MELalp	0.5	0.6	0.4
A	Distou	1.5	1.5	1.7	E	MELalp	0.4	0.6	0.5
					E	COPrig	0.4	0.4	0.7
B	Distou	1.3	0.9	1.5	E	MELalp	1.3	0.8	0.6
B	Distou	0.5	0.4	0.7	E	COPrig	0.4	0.3	0.5
B	Distou	1.0	1.0	0.9					
B	ROSrub	1.3	1.5	2.1	F	none			
B	ROSrub	0.5	0.4	0.8					
B	ROSrub	0.8	0.2	0.9	G	none			
B	ROSrub	0.7	0.6	1.1					
					H	MELalp	0.7	0.8	0.4
C	Distou	1.3	1.4	1.9	H	Distou	1.5	1.0	1.4
C	Distou	1.2	1.1	0.8	H	MELalp	0.4	0.3	0.4
C	COPrig	0.2	0.3	0.6					
					I	none			
D	MELalp	0.8	0.8	0.6					
D	MELalp	0.8	1.7	0.4	J	MELalp	0.2	0.2	0.4
D	COP pro	1.1	1.0	2.1					
D	MELalp	1.7	0.8	0.4	K	MELalp	0.3	0.4	0.4
D	MELalp	1.4	1.2	0.5					
D	MELalp	1.3	1.3	0.4	L	none			
D	COP pro	1.6	1.0	1.6					
D	MELalp	1.2	0.8	0.4	M	none			
D	COPrig	0.3	0.3	0.6					
					N	Distou	2.0	2.1	1.5

¹ Measured to one decimal place

Appendix 13A Shrub Biomass Harvest record sheet – discrete shrubs

NEW ZEALAND CARBON MONITORING SYSTEM
SHRUB BIOMASS HARVEST RECORD SHEET

CMS PLOT IDENTIFIER: AN157 MEASURED BY: Mushtik McKay, Tom Orduksen
DATE: 8 November 2002 RECORDED BY: Hazel Broadbent

Sample identifier		Shrub Dimensions (m)			Whole plant weight				Subsample weight ¹						
Shrub No.	Species	(Discrete shrubs only)			Fresh weight (kg) ²			Fresh weight (kg) ²			Dry weight (kg) ³				
		Width ¹	Width ²	Height ¹	Leafy shoot	Branch	Stem	Leafy shoot	Branch	Stem	Leafy shoot	Branch	Stem	TOTAL	
1.	MELalp	0.8	0.7	0.4											
2.	COPrig	0.9	0.8	1.3				3.675	2.163						
3.	DISTho	1.5	0.8	1.3				2.500	1.760						
4.	DISTho	2.3	2.1	1.7	18.0	5.7	6.0	29.7	20.98	1.25	1.275	1.675			
													1.575		0.966
													0.905	0.890	1.106

NOTES
① Shrub - clump
② Leafy shoots — 1/6 subsample, branches & stems — 1/2 subsample.

¹ Do not subsample shrubs with a total fresh weight of <1 kg.
² Weigh shrubs with a fresh weight >5 kg to the nearest 100 g, and plants or subsamples <5 kg to the nearest 25 g.
³ Measure to the nearest 0.1 m.

Appendix 14 Continuous Shrub Cover record sheet

NEW ZEALAND CARBON MONITORING SYSTEM
CONTINUOUS SHRUB COVER RECORD SHEET

CMS PLOT IDENTIFIER: Z174 MEASURED BY: Thomas Paul, Stu McKenzie
 DATE: 9 December 2002 RECORDED BY: Lara Nicholson
 METHOD USED TO DETERMINE AREA OF BIOMASS PLOT: Tape distance / Horizontal distance (Delete one)

DIMENSIONS OF BIOMASS HARVEST PLOT

	Long axis 1	Long axis 2	Short axis 1	Short axis 2
Tape distance (m)	4.00	4.00	1.00	1.00
Slope angle (°)	12.7	11.6	5.4	3.8
Horizontal distance (m)	3.90	3.92	1.00	1.00

NON-DESTRUCTIVE PLOT MEASUREMENTS

Permanent plot (0.04 ha)							
Subplot	Species	Height (m) ¹	% Cover	Subplot	Species	Height (m) ¹	% Cover
1	none	0	0	20	ULE _{ew}	1.4	95
2	none	0	0	21	ULE _{ew}	1.0	85
3	ULE _{ew}	0.7	25	22	none	0	0
4	none	0	0	23	none	0	0
5	ULE _{ew}	1.0	15	24	ULE _{ew}	3.4	95
6	ULE _{ew}	0.9	25	25	ULE _{ew}	4.0	85
7	ULE _{ew}	0.8	45				
8	ULE _{ew}	0.8	15	Biomass harvest plot (4 x 1 m)			
9	none	0	0	Subplot	Species	Height (m) ¹	% Cover
10	none	0	0	1	ULE _{ew}	0.7	25
11	ULE _{ew}	0.9	25	2	ULE _{ew}	0.5	15
12	ULE _{ew}	1.6	65	3	ULE _{ew}	0.9	35
13	ULE _{ew}	2.1	75	4	ULE _{ew}	1.2	55
14	ULE _{ew}	2.0	75	5	ULE _{ew}	1.1	95
15	ULE _{ew}	1.4	55	6	ULE _{ew}	1.1	95
16	ULE _{ew}	1.6	55	7	none	0	0
17	none	0	0	8	none	0	0
18	none	0	0	9	ULE _{ew}	0.8	85
19	ULE _{ew}	1.4	65	10	ULE _{ew}	1.0	55

NOTES

Biomass harvest plot 9.5m from corner D on a bearing of 125°.

¹ Maximum vertical height of woody vegetation (to nearest 0.1m)

Manawhenua
Landcare Research

Take a photocopy of this sheet and the last remeasurement into the field to reconcile problems for the following plot.

Survey	Stewart Island East 1979-80
Area	Chew Tobacco
Line/plot	28
Sampling date	1979
Plot type	Diameter

[illegible]

Send resolved problems to:
The Curator
National Vegetation Survey (NVS) databank
Landcare Research
PO Box 69, Lincoln 8152

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

